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Technology and the Changing Family: A Unified Model of Marriage, Divorce, Educational Attainment, and Married Female Labor-Force Participation[†]

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Marriage has declined since 1960, with the drop being more significant for noncollege-educated individuals versus college-educated ones. Divorce has increased, more so for the noncollege-educated. Additionally, positive assortative mating has risen. Income inequality among households has also widened. A unified model of marriage, divorce, educational attainment, and married female labor-force participation is developed and estimated to fit the postwar US data. Two underlying driving forces are considered: technological progress in the household sector and shifts in the wage structure. The analysis emphasizes the joint role that educational attainment, married female labor-force participation, and marital structure play in determining income inequality. (*JEL* D13, D31, D83, I20, J12, J16, O33)

The character of American households has changed dramatically since World War II. First, the fraction of married households has plunged, both because of a rise in the fraction of never-married households and an increase in the rate of divorce. The change has been most notable for noncollege-educated households. Second, there has been a rise in assortative mating. That is, people are more likely to marry someone of the same educational level today than in the past. Third, the fraction of college-educated men and women has increased substantially. This is especially true for women. Fourth, there has been a dramatic rise in labor-force participation by married women. Fifth, income inequality across households has widened significantly.[1]

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[†] Go to http://dx.doi.org/10.1257/mac.20130156 to visit the article page for additional materials and author disclosure statement(s) or to comment in the online discussion forum.

[1] These facts are detailed in Section II.

The goal of this paper is to develop a unified theory capable of explaining this array of facts. The model has three key ingredients. First, marriage and divorce decisions are formalized within the context of a search-theoretic paradigm. People match randomly and only marry if both parties agree. A divorce occurs when one party in a marriage favors single life over married life. A divorcee is free to remarry if the opportunity arises. The attractiveness of a mate depends on his/her ability and educational level, as well as on the love arising from the relationship. Second, all individuals make a choice about whether to go to college. They do this based on their ability and their psychic cost of going to school. Third, married households must decide whether the woman should work. This depends on the wage women will earn in the market and loss in utility the household incurs when she works. Labor at home is used in household production.

There are two exogenous driving forces in the analysis: technological progress in the home and shifts in the wage structure. Technological progress in the home reduces the labor needed in household production. This makes it easier for married women to work in the market. Moreover, with better technology in the home, the economies of scale associated with married life matter less. Hence, this force promotes a decline in marriage and an increase in divorce. Two shifts in wage structure are entertained: an increase in the return to education and a decline in the gender wage gap. A rise in the return to education entices more men and women to go to college. Shrinkage in the gender wage gap encourages labor-force participation by married women and makes singlehood more affordable for women.

The framework developed connects the induced shifts in the structure of households to the rise in income inequality. Consider the following thought experiment: Suppose husbands and wives work full-time and there is no gender wage gap. Then, random matching would reduce household income inequality. For this effect to be operational, though, married women must work. Now, an increase in positive assortative mating works to amplify income inequality. This effect will be stronger if women at the upper end of the income distribution work more than those at the lower end.

The unified framework developed here is matched with US data from 1960 using a minimum distance estimation strategy. The procedure targets a collection of stylized facts concerning educational attainment, marriage and divorce, and married female labor-force participation. The framework fits the data for 1960 well. The structural parameter values obtained also look reasonable, and are tightly estimated. The model predictions for 2005 are then compared with the corresponding US data. A slight retuning of a very limited number of parameters is then undertaken before the framework is used to decompose the shift in family structure into its underlying driving forces.

Both driving forces are quantitatively important for explaining the changes in family structure outlined previously. The findings suggest that technological progress in the household sector accounts for the majority of the rise in married female labor-force participation. The narrowing of the gender wage gap in wages plays a secondary role here, too. Technological progress in the household sector also has a conspicuous effect in explaining the fall in marriage and the rise in divorce. Changes in the structure of wages are important for the increase in assortative mating and educational attainment.

While the rise in the skill (college) premium is the root cause for widening household income inequality, shifts in family structure provide a very important amplification mechanism. An increase in the return to education entices more people in the right-hand side of the ability distribution to go to college, which makes household incomes more disperse. Positive assortative mating implies that a high- (low-) earning woman is more likely to be matched in marriage with a high- (low-) earning man and this, too, heightens inequality. For this latter effect to be operational, however, married women must work in the labor force. Hence, the rise in married female labor-force participation also plays a role in generating household income inequality.

After a brief literature review in Section I, the remainder of the paper flows as follows: Section II describes the main facts in detail. The model is presented in Section III. Section IV discusses the calibration/estimation procedure for 1960, and then Section V considers the model results for 2005. Section VI decomposes the effects of each of the exogenous forces at play. Section VII discusses the implications of the developed framework for household income inequality. Some concluding remarks are offered in Section VIII.

I. Relation to the Literature

The framework developed here resembles, in some aspects, Greenwood and Guner (2009) who study the fall in marriage and the rise in divorce. However, their model does not have heterogeneity with respect to education and ability. By adding this in the current framework, it is possible to study assortative mating and inequality. Another related paper is by Regalia and Ríos-Rull (2001), which was ahead of its time. While their model does feature heterogeneity in both men and women, the focus is on accounting for the rise in the number of single mothers, something left out of the current analysis. They stress market forces, such as a movement in the gender wage gap, as explaining this rise, but a mechanism for studying the rise in assortative mating appears to be absent.

Jacquemet and Robin (2012) estimate a search and matching model of the marriage market for the United States. Their analysis focuses on how female and male wages affect marriage probabilities and the share of the marital surplus received by partners. Given this goal, there is no need to include endogenous divorce or educational attainment in their model, which is central to the current paper. Eckstein and Lifshitz (2011) study the effect that different mechanisms (schooling, the gender wage gap, fertility, and marriage and divorce) had on the rise in female labor-force participation during the twentieth century. They find that up to 42 percent of the change is left unexplained. They attribute this residual component to improvements in household technology and changes in social norms. This is consistent with the story told in this paper.

Parts of the picture have been addressed before elsewhere. Greenwood, Seshadri, and Yorukoglu (2005) analyze the importance of technological progress in the home sector for making it more feasible for married women to enter into the labor market.[2]

[2] Independent empirical work by Cavalcanti and Tavares (2008) and Coen-Pirani, León, and Lugauer (2010) also suggests that labor-saving household products have increased married female labor supply. Adamopoulou

However, they do not study the changes in household structure or inequality, as done here. The interaction between inequality and positive assortative mating has also been noted by Fernàndez and Rogerson (2001) and Fernàndez, Guner, and Knowles (2005). Chiappori, Iyigun, and Weiss (2009) discuss how positive assortative mating provides a marriage market return for female educational investment, in addition to the traditional labor market one. The same effect is at play in the model developed here and, together with the rise in married female labor-force participation, is important to explain the rise in household income inequality. The current work studies the relationship between assortative mating and household income inequality within the context of a structural model, which takes into account the endogenous response of household decisions to shifts in the economic environment. A deconstruction of the structural model's amplification mechanism is undertaken to show how induced changes in family structure can contribute to income inequality.

Some different ways in which marriage and female labor supply decisions interact in the current framework have been pointed out in the literature. Neeman, Newman, and Olivetti (2008), for example, argue that college-educated working women can afford to be more selective in marriage and this may lead to more stable marriages. Such an outside option effect is also operational in the current framework. Gihleb and Lifshitz (2013) document that a married woman who is more educated than her husband is more likely to work. They analyze how changes in assortative mating can account for shifts in married female labor supply. In the current analysis, both assortative mating and female labor market participation are endogenously determined. This is done within an equilibrium framework that can be used to study household income inequality.

II. Facts

The shape of the American household has changed dramatically during the last 50 years. Some salient features of this transformation are as follows:

- *The Decline in Marriage*: The fraction of the population that has ever been married has fallen dramatically since 1960. At that time, about 85 percent of college-educated individuals and 92 percent of those who were noncollege-educated between the ages of 25 and 54 were married (or had been married)—see Figure 1. (Data sources for this and all other figures are provided in Appendix A.) Today, only 81 (79) percent are.[3] Note that the fall in the fraction of the population that is married is greatest for noncollege-educated people. Part of the decline in marriage is due to a delay in the age of marriage. Part is due to a rise in divorce. In 1960, the fraction of the population that was divorced, as measured by the ratio of the currently divorced to the ever-married population, was 5 percent for the noncollege-educated populace and 3 percent

(2014) shows that these products also have contributed to the rise of cohabitation. Advances in maternal medicine and pediatric care played a similar role, as has been noted by Albanesi and Olivetti (forthcoming).

[3] Redoing Figure 1 with currently married and currently divorced individuals as fractions of the total population (instead of ever-married individuals as a fraction of the total population and the currently divorced as a fraction of the ever-married population) delivers very similar patterns.

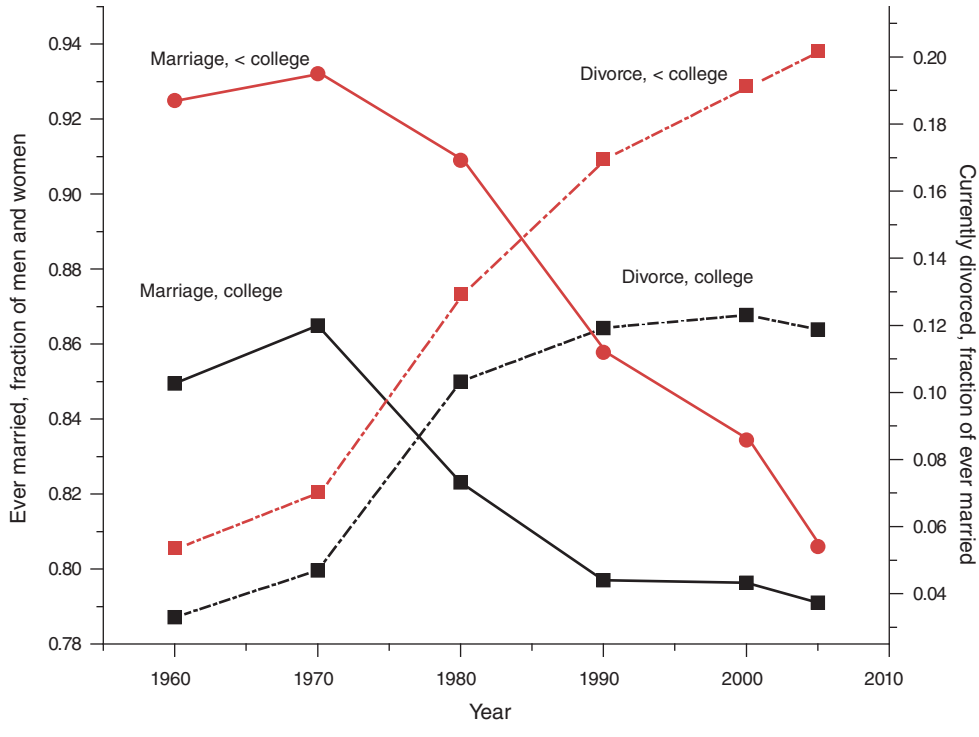


FIGURE 1. MARRIAGE AND DIVORCE BY EDUCATION

Source: See Appendix A for a description of the data used in all figures and tables.

for the college-educated segment. Today, it is around 20 percent for the former and 12 percent for the latter. Again, observe that divorce has risen more for the noncollege-educated vis-à-vis the college-educated population. The fact that the decline in marriage and the rise in divorce has affected college-educated and noncollege-educated people differentially has been noted both by sociologists Martin (2006) and economists Stevenson and Wolfers (2007).

- *The Rise in Assortative Mating*: When individuals marry today, as opposed to yesterday, they are more likely to pair with an individual from the same socioeconomic class. To see this, split the world into two socioeconomic classes, viz., noncollege-educated and college-educated, and compare the two contingency tables contained in Table 1.[4] The number in a cell in Table 1 shows the fraction of all matches that occur in the specified category. The figure in parentheses provides the fraction that would occur if matching occurred randomly. First, note that there is positive assortative mating. To see this, focus on the diagonal elements in the tables. These cells show the fraction of matches where husband and wife have the same educational levels. The difference between the actual and random matches in these cells is always positive, reflecting positive

[4] Greenwood et al. (2014) use five educational classes. The results there parallel the findings here for two classes.

TABLE 1—ASSORTATIVE MATING, AGES 25–54

1960			2005		
Husband	Wife		Husband	Wife	
	< College	College		< College	College
< College	0.855 (0.821)	0.023 (0.056)	< College	0.545 (0.427)	0.108 (0.226)
College	0.082 (0.115)	0.041 (0.008)	College	0.109 (0.227)	0.237 (0.120)

Statistics Measuring Assortative Mating

$\chi^2 = 33,451$	obs. $= 195,034$	$\chi^2 = 77,739$	obs. $= 288,423$
$\rho = 0.41$	$\delta = 1.08$	$\rho = 0.52$	$\delta = 1.43$

Source: See Appendix A for a description of the data used in all figures and tables.

assortative mating. The hypothesis of random matching is rejected by the χ^2 statistics.[5] The Pearson correlation coefficient, ρ, which measures the degree of association between the female and male educational categories, also is always positive. Second, the extent of positive assortative mating has become stronger over time. This can be seen in a number of ways. Note that between 1960 and 2005, the differences between the cells along the diagonals for the actual and random matrices increased. For each year, take the ratio of the traces of the matrices for actual and random marriages. Denote this ratio by δ, which divides the actual concordant matches by the random concordant ones. The higher this number is, the higher the degree of positive assortative mating. This ratio rises from 1.08 in 1960 to 1.43 in 2005. Additionally, the Pearson correlation coefficient, ρ, moves up from 0.41 to 0.52. To illustrate further the rise in assortative mating, consider running a regression for married couples of the form

$$
\begin{aligned}
(1)\quad \mathit{EDUCATION}_t^w = {} & \alpha + \beta \times \mathit{EDUCATION}_t^h \\
& + \sum_{y \in \mathcal{Y}} \gamma_t \times \mathit{EDUCATION}_t^h \times \mathit{DUMMY}_{y,t} \\
& + \sum_{y \in \mathcal{Y}} \theta_t \times \mathit{DUMMY}_{y,t} + \varepsilon_t, \quad \text{with } \varepsilon_t \sim N\left(0, \sigma^2\right),
\end{aligned}
$$

where: $\mathit{EDUCATION}_t^w \in \{0,1\}$ is the observed level of the wife's education in period t and takes a value of one if the woman completed college and a value of zero otherwise; $\mathit{EDUCATION}_t^h \in \{0,1\}$ is the husband's education; $\mathit{DUMMY}_{y,t}$ is a dummy variable for time such that $\mathit{DUMMY}_{y,t} = 1$, if $y = t$, and $\mathit{DUMMY}_{y,t} = 0$, if $y \neq t$; $t = 1960, 1970, 1980, 1990, 2000$, and 2005 represents the years in the sample and $\mathcal{Y}$ is the subset of these years that omits 1960. The coefficient γ_t measures the additional impact relative to 1960 that a husband's education will have on that of his wife. Note that the impact of a secular rise in female educational attainment is controlled for by the presence of the time dummy

[5] The χ^2 statistic is calculated as $\sum_{i=1}^{c} \sum_{j=1}^{r} \frac{(O_{i,j} - E_{i,j})^2}{E_{i,j}}$, where $O_{i,j}$ and $E_{i,j}$ are the observed and expected frequencies in cell (i,j). The degrees of freedom for the test are $(c-1)(r-1)$.

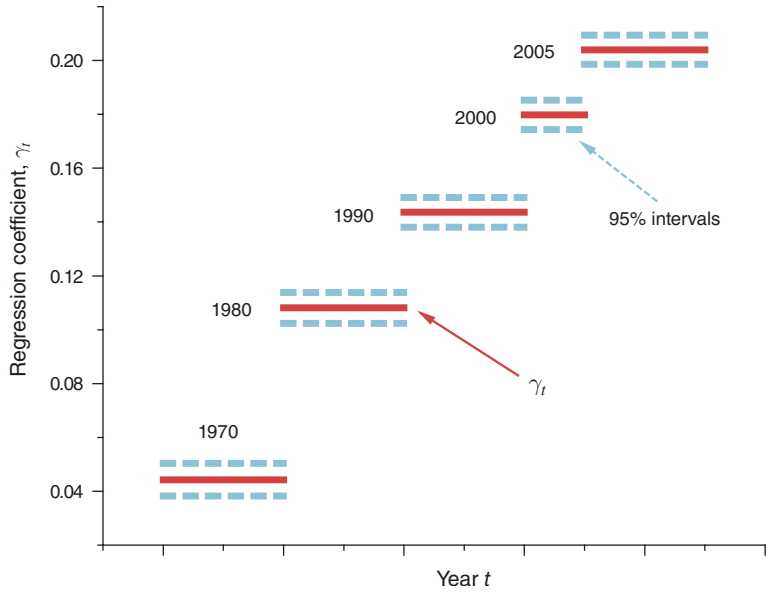


FIGURE 2. THE RISE IN ASSORTATIVE MATING

Notes: The solid line plots the regression coefficient, γ_t. The dashed lines show the 95 percent confidence intervals.

variable. So, how does γ_t change over time? Figure 2 plots the rise in the γ_ts.[6] The γ_t coefficients are significantly different from one another at the 95 percent confidence level. The same finding obtains if logits or probits are run instead. The rise in assortative mating has been noted before by sociologists Schwartz and Mare (2005).[7]

- *The Increase in Education and Labor-Force Participation by Married Women*: Labor-force participation by married women has increased dramatically over the last 50 years.[8] This is true for both college-educated and noncollege-educated women. In 1960, a minority of both classes of women worked. Now, the majority do—see Figure 3. At the same time, the number of women choosing to educate themselves has risen sharply. This may have been stimulated by a rise in the college premium, shown in Figure 4. College-educated women have always worked more than those without a college education. As female labor-force participation rose so did a married woman's contribution to family income—again, see Figure 3. Figure 4 also shows how the gender wage gap (the ratio of women's to men's wages) has narrowed.

[6] Appendix B connects the cells in the contingency tables in Table 1 with the coefficients on a regression of the form of (1). There is more on this in Section V.

[7] Blossfeld and Timm (2003) document that the rise is not just a US phenomenon, but it is also observed in other developed countries.

[8] Here, as discussed in Appendix A, labor-force participation is taken as the fraction of women who work (employment rate). Taking into account the unemployed women in the labor-force only changes these statistics slightly.

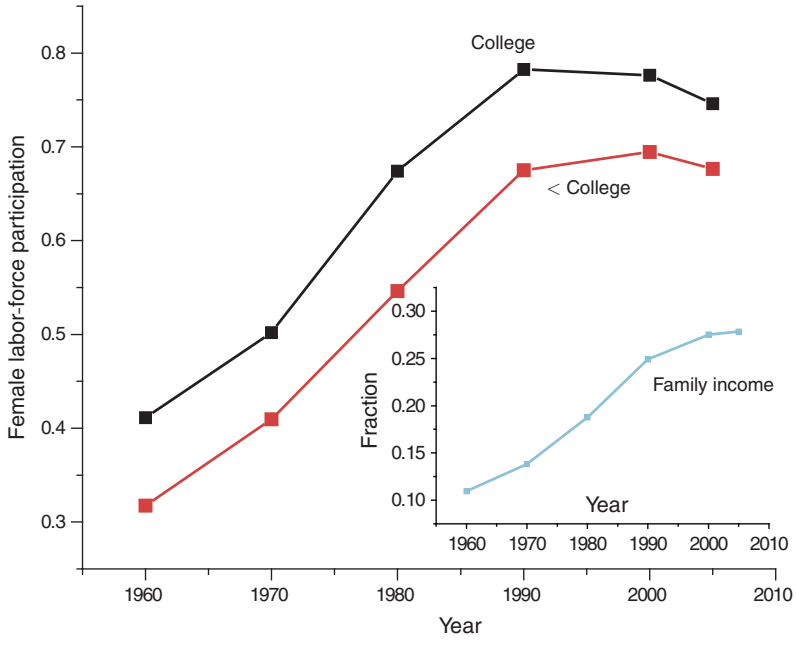


FIGURE 3. THE INCREASE IN MARRIED FEMALE LABOR-FORCE PARTICIPATION

Note: The inset panel shows the contribution of married women to family income.

- *The Increase in Household Income Inequality*: The distribution of income among households became more unequal between 1960 and 2005. The left panel of Figure 5 shows the Lorenz curves for 1960 and 2005. Lorenz curves plot the cumulative share of income at each income percentile against the cumulative percentile of households. If income was equally distributed among households, these curves would coincide with the 45° line. The Lorenz curves show that inequality increased. The Gini coefficient, which is twice the area between the Lorenz curves and the 45° line, increased from 0.31 to 0.43 between 1960 and 2005. Another way to see this is by plotting the household income relative to the mean household income in each percentile; this is done in the right panel of Figure 5. The relative income for all households below the eightieth percentile declined, while there was a significant increase for households that are at the top of the income distribution.

III. Model

What are the economic forces behind this dramatic shift in household characteristics? The idea can be described in a nutshell. People marry for both economic and noneconomic reasons: material well-being and love. On the material side of things, a woman's labor is important for both home production and market production. Over time the value of a woman's labor in home production has declined, due to technological progress in the household sector. Specifically, inputs into home production, such as dishwashers, frozen foods, microwave ovens, washing machines, and most

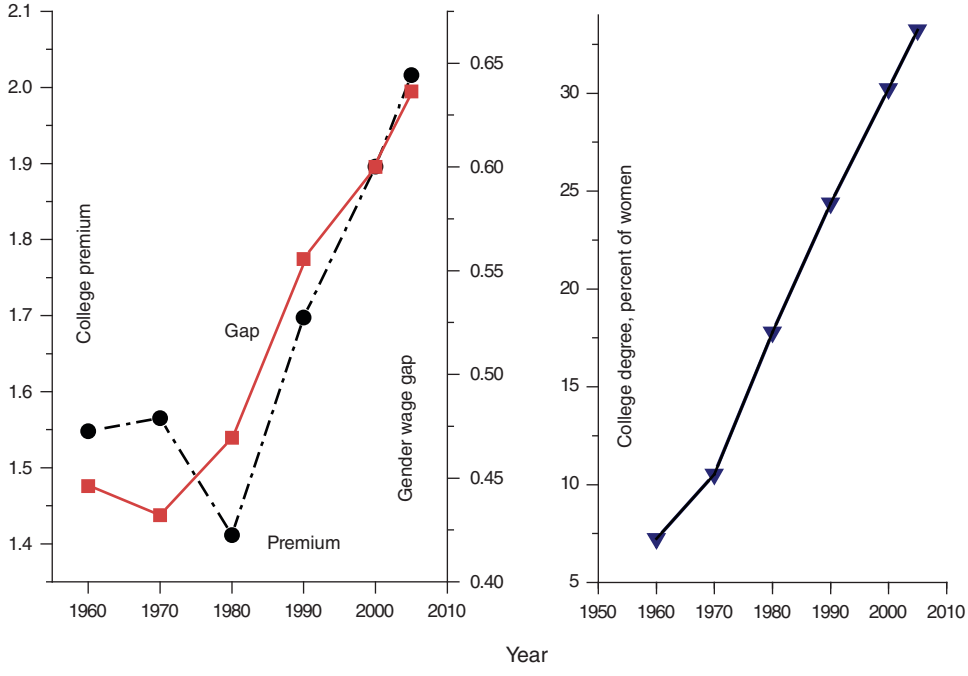


FIGURE 4. THE RISE IN FEMALE EDUCATIONAL ATTAINMENT, THE COLLEGE PREMIUM, AND THE NARROWING OF THE GENDER WAGE GAP

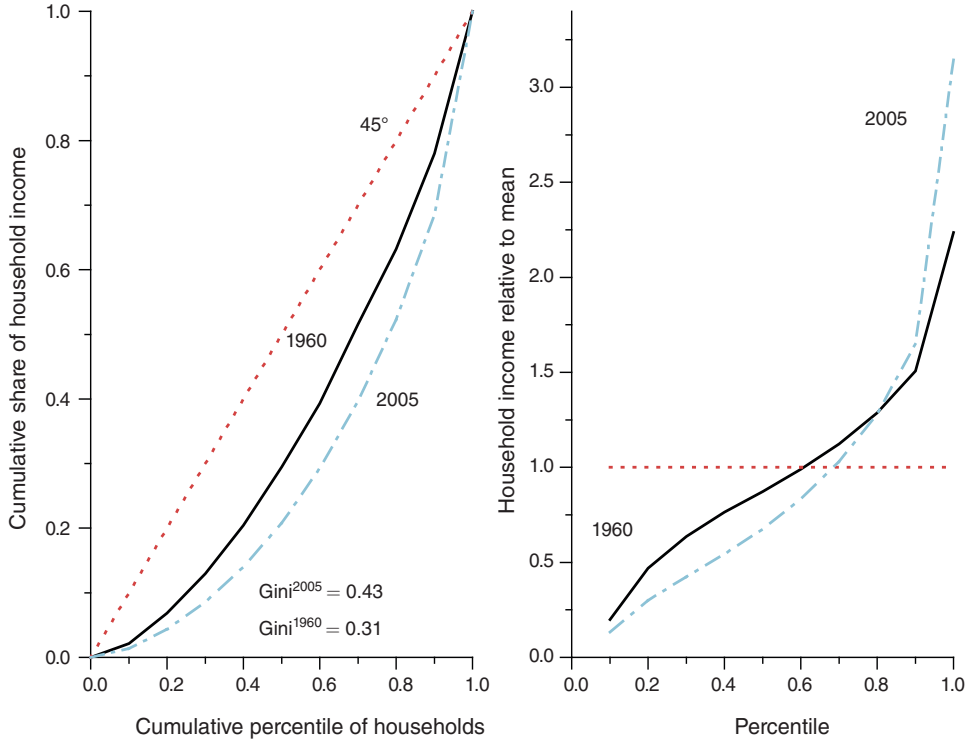


FIGURE 5. THE INCREASE IN HOUSEHOLD INCOME INEQUALITY

recently the internet, have reduced the need for household labor.[9] At the same time, the value of a woman's time in the market and her incentives to obtain additional education increased as a result of a narrower gender wage gap and a higher skill premium. Therefore, love and the value of a woman's labor on the market have come to play more important roles, relative to the value of a woman's labor in home production, in the decision about whether or not to get married and whom to marry.

A rise in the skill premium heightens income inequality. If more high-ability people go to college (relative to low-ability ones), then the earnings differential between high-ability and low-ability individuals will widen. A higher skill premium creates a greater incentive to match assortatively. So, changes in marriage patterns can intensify inequality. But, for this mechanism to have force, married women must work in the market. Otherwise, if women never worked, household income inequality would closely follow the inequality among men.

To formalize the discussion above, four things are required. First, a model of marriage and divorce is needed. Second, the framework must include a decision about whether or not married women should work. Third, the structure should incorporate an education decision. Fourth, people must be heterogenous in ability. This motivates the following setup.

A. *Setup*

Imagine an economy that is populated by equal numbers of men, m, and women, f. Some men and women are college-educated, while others are noncollege-educated. Some individuals of each gender will be married, the rest either divorced or never married. A person faces a constant probability of dying, δ, each period. Upon death an individual is replaced by a young doppelganger who is about to begin his or her adult life. A person enters adult life with an ability level $a \in A$. Initial ability is distributed across the population in line with the distribution function $A(a)$. It will be assumed that a is log normally distributed so that $\ln a \sim N\left(0, \sigma_a^2\right)$, where σ_a^2 denotes the variance of this zero-mean distribution.

The first decision that a young adult makes is whether to acquire an education. An uneducated man will earn the amount $w_0 a$ for each unit of labor supplied on the market, while an educated one earns $w_1 a$, where $w_1 > w_0$. A woman earns the fraction $\phi \in [0, 1]$ of what a comparable man does. This reflects the gender wage gap in labor income. Acquiring an education has an up-front utility cost κ. For a person of gender $g \in \{f, m\}$ with ability a, κ is a random variable drawn from the distribution $C_a^g(\kappa)$. Assume that $C_a^g(\kappa)$ is a normal distribution with mean η_g / a and variance σ_κ^2. The idea here is that the cost of learning is inversely related to a person's ability, so on average higher ability individuals have lower costs of education. There is, however, mixing, since even among individuals with high ability there will be some who draw a high cost of education. Let $e \in \mathcal{E} = \{0, 1\}$ represent whether $(e = 1)$ or

[9] While the focus here is on marriages, these forces reduce the need to live in large households in general. Bethencourt and Ríos-Rull (2009) and Salcedo, Schoellman, and Tertilt (2012) model the rise in single families, but in contexts not involving marriage. In a similar vein, Greenwood and Guner (2009) model the decisions of young people to leave their parent's home.

not ($e = 0$) a person has acquired an education. After the education decision, each individual will be characterized by an ability level, a, and an education level, e.[10] Denote a person's type by $(a, e) \in \mathcal{T} \equiv \mathcal{A} \times \mathcal{E}$.

Skill-biased technological progress results in skilled labor becoming more valuable relative to unskilled labor. Therefore, w_1 will grow over time relative to w_0 and the college premium moves up. As a consequence, more men will complete college. More women should finish college too. Take a single woman first. The income earned when single will now have risen for a college-educated woman, relative to a noncollege-educated one. Thus, a college-educated single female can now live better than before (again, relative to an noncollege-educated one). The extra income that a college education now provides means that a college-educated single woman can afford to be choosier when selecting a husband. The same reasoning applies to being single because of a divorce. Now, consider a married female. If she works, the return to a college education will have risen because her family will have more income (assuming that married women work). This provides an incentive to become more educated. This fact will also make a college-educated woman more attractive on the marriage market. The return from finding a better partner on the marriage market, in and of itself, may provide an extra incentive for women (and men) to invest in college. A decline in the gender wage gap (a rise in ϕ) will reinforce women's incentives to acquire a college education.

These forces should cause people to become pickier about their mate, causing a decline in marriage and a rise in divorce. Educated individuals are also less willing to marry uneducated agents, because with a higher skill premium the cost of marrying an uneducated person is higher. Hence, one would expect a rise in assortative mating. This mechanism intensifies the effect of a step up in the skill premium on income inequality.

At the beginning of each period, people must decide whether or not to work in the market during the period. Each person has one unit of time per period, which can be used for market or home production. Let h_f and h_m denote the hours worked by a woman and a man in the market, respectively. The workweek in the market is fixed. This is reflected in the two possible values that h can take, $h \in \mathcal{H} \equiv \{0, \overline{h}\}$. Suppose single agents always work full-time, allocating $\overline{h}$ to market and $1 - \overline{h}$ to household work. It is assumed that in marriage h is chosen only for the wife; the husband always works full-time.[11] Once a woman decides whether or not to get educated at the start of her life, her wage rate does not change. In particular, women who choose to stay home do not experience any future wage penalty. The importance of labor market experience for the labor supply decisions of married women is emphasized, among others, by Eckstein and Wolpin (1989) and Eckstein and Lifshitz (2011). Olivetti (2006) documents an increase in the returns to experience for women and links it to the rise in their market participation. If experience matters

[10] It is optimal for an individual to get an education in the first period. There is only a one-time utility cost, so by going to college early its benefits can be enjoyed for the longest possible horizon.

[11] The effect of changes in home technologies and wages on the time allocation decisions of husbands and wives has been analyzed by Bar and Leukhina (2011) and Knowles (2013).

for female wages, a higher risk of divorce can encourage wives to work, as discussed by Fernàndez and Wong (2014).

Home goods are produced according to

$$n = \left[\theta d^{\lambda} + (1-\theta)(z-h_T)^{\lambda}\right]^{1/\lambda}, 0 < \lambda < 1, \tag{2}$$

where d is the amount of household durables, h_T is the total amount of time spent on market work, and $z \in \{1, 2\}$ is the household's size. The restriction that $0 < \lambda < 1$ implies that household durables, d, and time, $z - h_T$, are substitutes in household production. Household durables, d, can be purchased at the price p in terms of market goods. The substitutability between labor and durable goods in household production implies that labor will be released from married households if the price of durables drops because of technological advance in the home sector. This promotes a rise in married female labor-force participation.

At the end of each period, a single person will meet someone else of the opposite sex, with ability level a^* and education e^*. The couple will then draw two shocks. The first is a match-specific bliss shock $b \in \mathcal{B}$, taken from the distribution $F(b)$. In particular, b will be normally distributed so that $b \sim N\left(\bar{b}_s, \sigma^2_{b,s}\right)$, where $\bar{b}_s$ and $\sigma^2_{b,s}$ denote the mean and variance of the bliss distribution that an unmarried couple draws from. In a marriage, the bliss shock evolves according to the distribution $G(b'|b)$. Specifically, the bliss shock is assumed to follow the autoregressive process $b' = (1-\rho_{b,m})\bar{b}_m + \rho_{b,m} b + \sigma_{b,m}\sqrt{1-\rho^2_{b,m}}\varepsilon$, with $\varepsilon \sim N(0,1)$. Here $\bar{b}_m$ and $\sigma^2_{b,m}$ represent the long-run mean and variance of this process, while $\rho_{b,m}$ is the coefficient of autocorrelation. A married person will decide whether to remain with his/her current partner partly on the value of this bliss shock.

The second shock, $q \in \mathcal{Q}^e \equiv \{q^e_l, q^e_h\}$, measures the cost for a married woman of going to work.[12] The two-point set, $\mathcal{Q}^e$, that the shock, q, is drawn from depends on the education level of the husband; i.e., there is one distribution for couples with college-educated husbands and another one for couples with noncollege-educated husbands. This assumption is elaborated on further when the estimation strategy is discussed later. Without loss of generality, assume that $q^e_l < q^e_h$ and that a matched couple draws, before their marriage decision, q^e_l and q^e_h with equal probabilities. This shock is assumed to be permanent and hence does not change over time.[13] Some prospective families may place a greater value on the woman staying at home; perhaps they are more likely to have children, a factor abstracted away from here. After drawing the shock, the couple then decides whether to marry. This decision will be based upon both economic and noneconomic considerations, as will soon become clear. The discrete distribution for q is represented by $Q^e(q)$.

One barrier for married women going to work is the presence of young children. Modeling fertility endogenously is a substantial complication. The unitary model of the household must now be abandoned because the presence of children affects men and women differently upon a divorce. Some form of a bargaining model must

[12] Guner, Kaygusuz, and Ventura (2012) employ a similar strategy to model female labor-force participation.

[13] It is assumed that this cost has no bite once a marriage is dissolved. As a result, and absent an explicit fertility decision or a cost of divorce, divorced and never-married individuals are indistinguishable in the model economy.

now be used—see Greenwood, Guner, and Knowles (2003). As a practical matter, an accounting decomposition exercise along the lines of Greenwood et al. (2014) shows that changing fertility has little impact on income inequality.[14] Part of the cost of a married woman going to work might be child care costs, so q could partially reflect these. The effect of these latter costs on married female labor supply is examined by Attanasio, Low, and Sánchez-Marcos (2008).

The noneconomic factors underlying a marriage consist of the value of b, the value of q, and a measure of the compatibility for a couple. For a couple with education levels e and e^*, this compatibility is represented by the function $M(e, e^*)$, where

$$M(e, e^*) = \mu_0(1 - e)(1 - e^*) + \mu_1(ee^*).$$

If neither person went to college, then this function returns a value of μ_0, since $e = e^* = 0$, while if both are college-educated then it gives a value of μ_1. It yields 0 for all other cases.[15] If these parameters do not change over time, then any changes in assortative mating over time will be generated endogenously by the model only in response to technological progress in the household sector and to changes in the wage structure. Changes in μ_0 and μ_1, on the other hand, can capture changes in assortative mating because of other factors, such as changing social norms in the marriage market.[16] The economic factors underpinning a marriage are based upon each person's ability and educational attainment; that is, the (a, e, a^*, e^*) combination.

Now, suppose married women stay at home when the skill premium rises. It is still possible for more women to go to college. The increased return to skill will entice more men to acquire a college education. The fact that there are more college-educated men around implies that there may be a bigger incentive for women to invest in a college education in order to become more desirable on the marriage market (because of compatibility considerations).

Last, let all people discount the future at the rate $\beta = \tilde{\beta}(1 - \delta)$, where $\tilde{\beta}$ is the subjective discount factor. Suppose that for singles, tastes over the consumption of market goods, c, and nonmarket ones, n, are represented by

$$T_s(c, n) = \frac{1}{1 - \zeta}(c - \mathfrak{c})^{1-\zeta} + \frac{\alpha}{1 - \xi} n^{1-\xi},$$

where $\mathfrak{c}$ is a fixed cost in terms of market goods. Assume that in marriage the utility derived from consumption and love is a public good. Momentary utility for a married household is

$$T_m(c, n) = \frac{1}{1 - \zeta}\left(\frac{c - \mathfrak{c}}{1 + \chi}\right)^{1-\zeta} + \frac{\alpha}{1 - \xi}\left(\frac{n}{1 + \chi}\right)^{1-\xi},$$

[14] To be more precise, in such an accounting exercise, imposing the 1960 fertility patterns on the 2005 economy only increases the Gini coefficient from 0.430 to 0.434.

[15] It could alternatively be assumed that a fraction of agents match within their own education group while the remaining agents match randomly. Here assortative mating would be exogenous.

[16] Modeling changes in societal norms, a factor out of the purview of the current analysis, is the subject of Fernandez, Fogli, and Olivetti (2004).

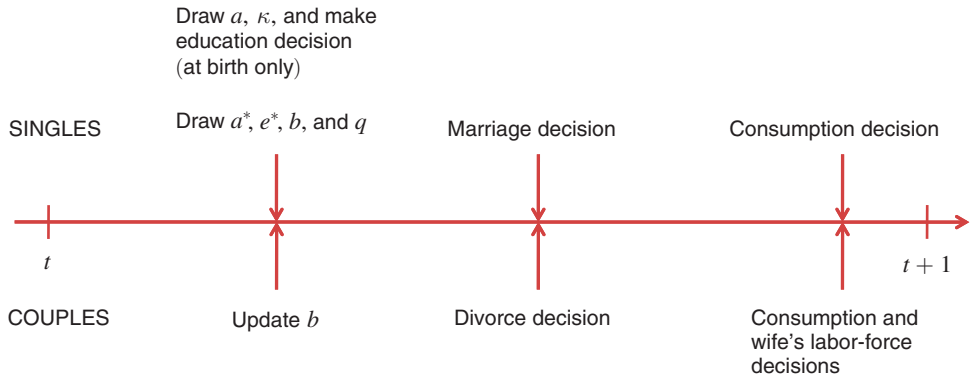


FIGURE 6. TIMING OF DECISIONS

where $\chi < 1$ is the household equivalence scale. The equivalence scale reflects the fact that there are economies of scale in household consumption so that a two-person household requires less than twice the consumption of a one-person household to realize the same level of utility as the latter. The variables $\mathfrak{c}$ and χ provide an economic motive for marriage. A two-person household will be better off than a single-person one. As incomes grow over time, the fixed cost, $\mathfrak{c}$, will be easier to cover. Therefore, a trend toward smaller households will emerge. This will be reflected in a lower marriage rate and a higher divorce rate.

Now, suppose that $\xi > \zeta$, which implies higher diminishing marginal utility for household goods vis-à-vis market ones. In this case, single households will benefit the most from technological advance in the home sector. This is because *at the margin* they will be the most intensive users of home production, as paradoxical as this may seem. That is, while the economically better off married couple (due to economies of scale) will consume more of all goods, relative to a single person, they will not consume twice as much home goods, because they will prefer to direct, at the margin, their larger consumption bundle toward market ones. Technological progress in the home allows for more home goods to be produced. It will improve single life the most because the marginal value for a home-produced good is highest for singles. This operates to reduce household size over time.

To complete the description of the setting, the timing of events within a period is illustrated in Figure 6. At any point, the model economy will be populated by married, single-male, and single-female households. Some of these married households will have husbands and wives who are college-educated, while others will have two noncollege-educated members, and yet others will have a college-educated husband and a noncollege-educated wife or vice versa. Similarly, single households will also differ by their educational attainments. Furthermore, not all educated agents will have the same earnings since they have different ability levels. Finally, some married women will participate in the labor market while others won't. These differences will generate inequality among households, and the model economy provides a natural framework to study how changes in household structure affect inequality.

B. *Singles*

Consider the consumption decision facing a single. This is a purely static problem. For a single person of gender $g \in \{f, m\}$ with ability a and educational attainment $e \in \{0, 1\}$, the problem is given by

$$U_s^g(a, e) \equiv \max_{c,n,d} T_s(c, n), \tag{3}$$

subject to

$$c = \begin{cases} w_e \phi a \bar{h} - pd, & \text{if } g = f, \\ w_e a \bar{h} - pd, & \text{if } g = m, \end{cases}$$

and

$$n = \left[\theta d^{\lambda} + (1 - \theta)\left(1 - \bar{h}\right)^{\lambda}\right]^{1/\lambda}.$$

Next, turn to the marriage decision. Consider a single person of gender $g \in \{f, m\}$ with ability a and educational attainment e. Suppose that this individual meets someone of the opposite gender, g^*, who has ability a^* and educational attainment e^* and the potential couple draws shocks b and q. Will they get married? To answer this question, let $V_s^g(a, e)$ and $V_s^{g^*}(a^*, e^*)$ represent the expected lifetime utilities that both parties will realize if they remain single in the current period. Likewise, denote the expected lifetime utility that is associated with a marriage in the current period by $V_m^g(a, e, a^*, e^*, b, q)$. A marriage will occur if and only if

$$V_m^g(a, e, a^*, e^*, b, q) \geq V_s^g(a, e) \quad \text{and} \quad V_m^{g^*}(a^*, e^*, a, e, b, q) \geq V_s^{g^*}(a^*, e^*). \tag{4}$$

Observe that, for a marriage to happen, it must be the first choice for *both* parties. Let the indicator function $\mathbf{1}^g(a, e, a^*, e^*, b, q)$ take a value of 1, if both people in the match want to marry, and value of zero otherwise. Thus,

$$\mathbf{1}^g(a, e, a^*, e^*, b, q) = \begin{cases} 1, & \text{if (4) holds,} \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

[Observe that $\mathbf{1}^g(a, e, a^*, e^*, b, q) = \mathbf{1}^{g^*}(a^*, e^*, a, e, b, q)$.]

The value of being single in the current period will depend on the distribution of potential future mates on the marriage market. Each mate is indexed by their (a^*, e^*) combination. Let the distribution of potential mates from the opposite gender be represented by $\hat{\mathbf{S}}^{g^*}(a^*, e^*)$. This will be elaborated on later. Define the variable $x(g)$ by

$$x(g) = \begin{cases} e, & \text{if } g = m, \\ e^*, & \text{if } g = f. \end{cases} \tag{6}$$

The value function for a single person of gender g with ability a and educational attainment e can now be expressed as

$$(7)\quad V_s^g(a,e) = U_s^g(a,e) + \beta \int_{\mathcal{B}}\int_{\mathcal{T}}\int_{\mathcal{Q}^{x(g)}} \Big\{ \mathbf{1}^g(a,e,a^*,e^*,b,q) V_m^g(a,e,a^*,e^*,b,q) + \big[1 - \mathbf{1}^g(a,e,a^*,e^*,b,q)\big] V_s^g(a,e) \Big\} dQ^{x(g)}(q) d\hat{\mathbf{S}}^{g^*}(a^*,e^*) dF(b),$$

$$\text{for } g = f,m.$$

Embedded in the above dynamic programming problem is the assumption that one will draw a mate next period with an ability level less than a^* and education level e^* with probability $\hat{\mathbf{S}}^{g^*}(a^*, e^*)$.[17] Note there is a slight asymmetry in the form of the value functions for single men and women due to presence of the function $x(g)$, which captures the cost of a married woman going to work as a function of her husband's education level.

C. *Couples*

The static consumption problem for a married couple is

$$(8)\qquad U_m^g(a,e,a^*,e^*,q) \equiv \max_{c,n,d,h^f \in \{0,1\}} T_m(c,n) - h^f q,$$

subject to

$$c = \begin{cases} w_{e^*} a^* \bar{h} + w_e \phi a \bar{h} h^f - pd, & \text{if } g = f, \\ w_e a \bar{h} + w_{e^*} \phi a^* \bar{h} h^f - pd, & \text{if } g = m, \end{cases}$$

and

$$n = \left[\theta d^\lambda + (1-\theta)\big(2 - \bar{h} - \bar{h} h^f\big)^\lambda\right]^{1/\lambda}.$$

Recall that all utility flows are public goods within a marriage. So, the couple picks c, n, d, and h^f together. Working in the market takes away the fraction $\bar{h}$ of a person's time endowment. Recall that husbands are assumed to work full-time. The variable $h^f \in \{0,1\}$ represents the wife's labor-force participation. It takes a value of 1 when the woman works and a value of 0 if she doesn't. Once again, the variable q gives the cost of going to work for a married woman. This cost is netted out of household utility when the woman works. Let $H^f(a,e,a^*,e^*,q) \in \{0,1\}$ denote the female labor-force participation decision for a couple of type (a,e,a^*,e^*,q).

A divorce will occur if and only if

$$(9)\quad V_s^g(a,e) > V_m^g(a,e,a^*,e^*,b,q) \quad \text{or} \quad V_s^{g^*}(a^*,e^*) > V_m^{g^*}(a^*,e^*,a,e,b,q).$$

[17] Other matching processes could be envisaged, such as the Gale and Shapley algorithm employed by Del Boca and Flinn (2014).

Therefore, the indicator function $\mathbf{1}^g(a, e, a^*, e^*, b, q)$, specified by equation (5), will return a value of one, if both the husband and wife want to remain married, and will give a value of zero, if one of them desires a divorce. Given this, the value function for a married person reads

$$
\begin{aligned}
(10)\quad V_m^g(a,e,a^*,e^*,b,q) \;=\; & U_m^g(a,e,a^*,e^*,q) + b + M(e,e^*) \\
& + \beta\Big\{\int_{\mathcal{B}}[\mathbf{1}^g(a,e,a^*,e^*,b',q)V_m^g(a,e,a^*,e^*,b',q) \\
& + [1 - \mathbf{1}^g(a,e,a^*,e^*,b',q)]V_s^g(a,e)]dG(b'|b)\Big\},
\end{aligned}
$$

for $g = f, m$.

This value function is used in equations (4), (5), (7), and (9); likewise, (7) is employed in equations (4), (5), (9), and (10).

D. *Educational Choice*

People choose their education level at the beginning of adult life after they observe κ, the utility cost of education. The problem they face is

$$
(11)\qquad \max_{e\in\{0,1\}} \{V_s^g(a,e) - e\kappa\},
$$

where V_s^g is defined by equation (7). The decision rule stemming from this problem will be represented by a simple threshold rule, since $V_s^g(a,1) > V_s^g(a,0)$,

$$
(12)\qquad E_a^g(\kappa) \;=\; \begin{cases} 1, & \text{if } \kappa \le \tilde{\kappa}_a^g \equiv V_s^g(a,1) - V_s^g(a,0), \\ 0, & \text{if } \kappa > \tilde{\kappa}_a^g. \end{cases}
$$

The total number of agents of gender g with ability a who choose to get a college degree is then given by

$$
\int_{-\infty}^{\infty} E_a^g(\kappa)dC_a^g(\kappa),
$$

and the total number of gender g agents with a college education is

$$
\int_0^{\infty}\int_{-\infty}^{\infty} E_a^g(\kappa)dC_a^g(\kappa)dA(a).
$$

E. *Steady-State Equilibrium*

The dynamic programming problem for a single person, or equation (7), depends upon knowing the solution to the problem for a married person, as given by (10), and vice versa. Furthermore, to solve the single's problem requires knowing the

steady-state distribution of potential mates in the marriage market, $\mathbf{S}^g(a)$. The non-normalized steady-state distribution for singles is

$$(13)\quad \mathbf{S}^g(a',e')$$

$$= (1-\delta)\int_{\mathcal{B}}\int_{\mathcal{T}}\int_{\mathcal{T}}^{a',e'}\int_{\mathcal{Q}^{x(g)}}\left[1-\mathbf{1}^g(a,e,a^*,e^*,b,q)\right]dQ^{x(g)}(q)d\mathbf{S}^g(a,e)d\hat{\mathbf{S}}^{g^*}(a^*,e^*)dF(b)$$

$$+ (1-\delta)\int_{\mathcal{B}}\int_{\mathcal{B}}\int_{\mathcal{T}}\int_{\mathcal{T}}^{a',e'}\int_{\mathcal{Q}^{x(g)}}\left[1-\mathbf{1}^g(a,e,a^*,e^*,b,q)\right]d\mathbf{M}^g(a,e,a^*,e^*,b_{-1},q)dG(b|b_{-1})$$

$$+ \delta e'\int_0^{a'}\int_{-\infty}^{\infty}E_a^g(\kappa)dC_a^g(\kappa)dA(a) + \delta(1-e')\int_0^{a'}\int_{-\infty}^{\infty}\left[1-E_a^g(\kappa)\right]dC_a^g(\kappa)dA(a), \text{ for } g=f,m,$$

where again $x(g)$ is defined by (6). In the above recursion, $\mathbf{M}^g(a,e,a^*,e^*,b_{-1},q)$ represents the steady-state distribution over married people, and $\hat{\mathbf{S}}^{g^*}(a^*,e^*)$ denotes the normalized distribution for singles of the opposite gender and is defined by

$$(14)\qquad \hat{\mathbf{S}}^{g^*}(a^*,e^*) \equiv \frac{\mathbf{S}^{g^*}(a^*,e^*)}{\int_{\mathcal{T}} d\mathbf{S}^{g^*}(a^*,e^*)}.$$

The first term in equation (13) counts those singles who failed to match in the current period. The second term enumerates the flow into the pool of singles from failed marriages. The last two terms represent the arrival of new adults (the doppelgangers).

In similar fashion, the distribution of married men and women is defined by

$$(15)\quad \mathbf{M}^g(a',e',a^{*\prime},e^{*\prime},b',q')$$

$$= (1-\delta)\int_{\mathcal{B}}^{b'}\int_{\mathcal{T}}^{a',e'}\int_{\mathcal{T}}^{a^{*\prime},e^{*\prime}}\int_{\mathcal{Q}^{x(g)}}^{q'}\mathbf{1}^g(a,e,a^*,e^*,b,q)dQ^{x(g)}(q)d\hat{\mathbf{S}}^{g^*}(a^*,e^*)d\mathbf{S}^g(a,e)dF(b)$$

$$+ (1-\delta)\int_{\mathcal{B}}^{b'}\int_{\mathcal{B}}\int_{\mathcal{T}}^{a',e'}\int_{\mathcal{T}}^{a^{*\prime},e^{*\prime}}\int_{\mathcal{Q}^{x(g)}}^{q'}\mathbf{1}^g(a,e,a^*,e^*,b,q)d\mathbf{M}^g(a,e,a^*,e^*,b_{-1},q)dG(b|b_{-1}),$$

$$\text{for } g=f,m.$$

The first term on the right-hand side measures the flow into marriage from single life. Only $1-\delta$ of these matches will last into the next period. The second term counts the number of marriages that will survive from the current period into the next one. Computing a steady-state solution for the model amounts to solving a fixed-point problem, as the following definition of equilibrium should make clear. Note that $\mathbf{M}^g(a,e,a^*,e^*,b_{-1},q) = \mathbf{M}^{g^*}(a^*,e^*,a,e,b_{-1},q)$.

DEFINITION 1: *A stationary matching equilibrium is a set of value functions for singles and marrieds, $V_s^g(a,e)$ and $V_m^g(a,e,a^*,e^*,b,q)$; an education decision rule for*

singles, $E_a^g(\kappa)$*; a matching rule for singles and married couples*, $\mathbf{1}^g(a,e,a^*,e^*,b,q)$*; and stationary distributions for singles and married couples*, $\mathbf{S}^g(a,e)$ *and* $\mathbf{M}^g(a,e,a^*,e^*,b_{-1},q)$*, all for* $g = f,m$*, such that:*

(*i*) *The value function* $V_s^g(a,e)$ *solves the single's recursion* (7)*, taking as given her/his indirect utility function,* $U_s^g(a,e)$*, from problem* (3)*, the value function for a married person,* $V_m^g(a,e,a^*,e^*,b,q)$*, the matching rule for singles,* $\mathbf{1}^g(a,e,a^*,e^*,b,q)$*, and the normalized distribution for singles,* $\hat{\mathbf{S}}^g(a,e)$*, defined by equation* (14).

(*ii*) *The value function* $V_m^g(a,e,a^*,e^*,b,q)$ *solves a married person's recursion* (10)*, taking as given her/his indirect utility function,* $U_m^g(a,e,a^*,e^*,q)$*, from problem* (8)*, the matching rule for a married couple,* $\mathbf{1}^g(a,e,a^*,e^*,b',q)$*, and the value function for a single,* $V_s^g(a,e)$.

(*iii*) *The decision rule* $E_a^g(\kappa)$ *solves a single's education problem* (11)*, taking as given* $V_s^g(a,e)$ *from equation* (7).

(*iv*) *The matching rule* $\mathbf{1}^g(a,e,a^*,e^*,b,q)$ *is determined in line with equation* (5)*, taking as given the value functions* $V_s^g(a,e)$ *and* $V_m^g(a,e,a^*,e^*,b,q)$.

(*v*) *The stationary distributions* $\mathbf{S}^g(a,e)$ and $\mathbf{M}^g(a,e,a^*,e^*,b_{-1},q)$ *solve equations* (13) *and* (15)*, taking as given the decision rule for an education,* $E_a^g(\kappa)$*, and the matching rule* $\mathbf{1}^g(a,e,a^*,e^*,b,q)$.

IV. Fitting the Model to the United States in 1960

The model developed will now be fit to the US data for 1960. There are many parameters. A few of them are easy to choose and can be assigned on the basis of a priori information. Most of the parameters, however, will be fitted using a minimum distance estimation procedure. The estimation procedure will focus on the 1960 US economy. In Section V the model will be simulated using 2005 wages and durable goods prices and the resulting fit will be examined. It will be assumed that the model is in a steady state for each of these years.

A. *A Priori Information*

The easy ones are done first. The length of a period is one year. Let $\tilde{\beta}$ (the subjective discount factor) be 0.96, a standard value in macroeconomic studies, such as in Prescott (1986). All the targets for the estimation are calculated for individuals between ages 25 and 54, which corresponds to an operational lifespan of 30 years. Set $\delta = 1/30 = 0.033$, so that individuals in the model also live 30 years on average. This would dictate a value for the discount factor of $\beta = 0.960 \times (1 - 0.033)$. Assigning a value for the workweek, $\overline{h}$, is straightforward. Assume a 40-hour workweek. Since there are 112 nonsleeping hours in a week,

let $\bar{h} = 40/112 = 0.36$.[18] Last, the household production parameters, θ and λ, have been estimated by McGrattan, Rogerson, and Wright (1997). Their numbers, $\theta = 0.21$ and $\lambda = 0.19$, are used here.[19] Finally, in line with the Organization for Economic Co-operation and Development (OECD) equivalence scale, set $\chi = 0.70$. To summarize, the parameter values picked on the basis of a priori information are displayed in Table 2.

B. *Minimum Distance Estimation*

This leaves 23 parameters to be assigned. There are six preference parameters, $\{\zeta, \mathfrak{c}, \alpha, \xi, \mu_0, \mu_1\}$; five parameters for the marital bliss shocks, $\{\bar{b}_s, \sigma_{b,s}, \bar{b}_m, \sigma_{b,m}, \rho_{b,m}\}$; three wage parameters $\{w_{0,1960}, w_{1,1960}, \phi_{1960}\}$; one parameter for durable goods prices, p_{1960}; three parameters for the cost of education, $\{\eta_f, \eta_m, \sigma_\kappa\}$; and one parameter for the ability distribution, σ_a. It is assumed that q_h and q_l differ by the education level of the husband. Let q_h^1 and q_l^1 denote the cost of joint work for couples with a college-educated husband, and q_h^0 and q_l^0 be the corresponding values for households with a noncollege-educated husband. This adds four more parameters. For both types of husbands, it is assumed that there is an equal chance of drawing a high or a low cost. Normalize the wage rate for a noncollege-educated man in 1960 to be one, so that $w_{0,1960} = 1$. The remaining 22 parameters are estimated so that the model matches, as closely as is possible, a set of 24 data moments for 1960.[20]

The data targets are as follows:

- *Educational Attainment*: The fraction of men and women who went to college.
- *Vital Statistics*: The fraction of the population that has ever-been married by educational level, and that is currently divorced (out of the ever-married populace) by education level.[21]

[18] An alternative would be to set $\bar{h}$ to actual hours worked per week. The value of $\bar{h}$ would then be 0.37, 0.35, and 0.39 in 1960 for single men, single women, and married men, respectively, and 0.38, 0.36, and 0.40 in 2005. Simulating the model economy for 1960 and 2005 with these values, instead of $\bar{h} = 0.36$, produces almost identical results.

[19] The parameter λ determines the elasticity of substitution between durable goods and household time, $1/(1 - \lambda)$. McGrattan, Rogerson, and Wright (1997) identify this parameter using time series variation. Since targets from a single year (specifically, 1960) are used to estimate the parameters here, λ is not included among them. The online Appendix displays the 1960 model statistics when λ is increased or decreased by 20 percent, while all other parameters are kept at their benchmark values. Changes in λ do not have any major effect on 1960 targets.

[20] In the data used, observations come from a mixture of different cohorts. In the model there is essentially a single infinite horizon cohort, with some of its members dying each period and being replaced with young doppelgangers. One way to get the data to be close to the steady-state approximation is to use averages of a subperiod rather than just a single year. The decennial census that is used to compute the moments contains data for 1960 only. Computing the same data targets using the Current Population Survey (CPS) for several years in the 1960s (1962–1965) yields remarkably similar statistics.

[21] In the model economy, individuals form households both because of love and to enjoy economies of scale in household production and consumption. Hence, it is a model of couples living together rather than being legally married. While it is possible to combine the married and cohabiting population to arrive at a stock of people who live together, it is more problematic to calculate a separation rate for cohabiting people. In the US census, the divorced category only covers those who had been married in the past. See Gemici and Laufer (2014) for a study of cohabitation and marriage. These authors calculate dissolution rates for married and cohabiting couples from the Panel Study of Income Dynamics. The calculation of such rates, however, is only possible after 1978.

TABLE 2—PARAMETERS: A PRIORI INFORMATION

Category	Parameter values	Criteria
Preferences	$\chi = 0.70$	OECD scale
	$\tilde{\beta} = 0.96$	Prescott (1986)
Household technology	$\theta = 0.21, \lambda = 0.19$	McGrattan et al. (1997)
Death probability	$\delta = 1/30$	A 30-year lifespan
Hours	$\bar{h} = 0.36$	Data

- *Assortative Mating*: A contingency table for marriage that contains the fractions of marriages for each possible combination of educational levels for both the husband and wife.
- *Married Female Labor-Force Participation*: The fraction of married women, classified by the education levels of husbands and wives, that work, and the share of household income provided by wives.
- *Skill Premium and Gender Wage Gap*: The earnings ratio between college-educated and noncollege-educated men (the skill premium), and the earnings ratio between women and men (the gender wage gap).
- *Inequality*: The Gini coefficient for earnings inequality among households; the 90-to-10 and 90-to-50 percentile ratios; income inequality across married households by the educational attainments of husbands and wives; and the ratio of single female to married household income.

Before the parameter estimates and the model fit are presented, a comment on the skill premium and gender wage gap as targets is in order. Take the skill premium first. Wages are needed for noncollege and college-educated men in 1960; viz., $w_{0,1960}$ and $w_{1,1960}$. Recall that $w_{0,1960} = 1$. The college premium in 1960 for the model is the average ratio of the earnings for a college-educated man to a noncollege-educated one, as given by

$$\frac{w_{1,1960}\left[\int_0^\infty\int_{-\infty}^\infty aE_a^m(\kappa)dC_a^m(\kappa)dA(a)\right]/\left[\int_0^\infty\int_{-\infty}^\infty E_a^m(\kappa)dC_a^m(\kappa)dA(a)\right]}{\left\{\int_0^\infty\int_{-\infty}^\infty a[1-E_a^m(\kappa)]dC_a^m(\kappa)dA(a)\right\}/\left\{\int_0^\infty\int_{-\infty}^\infty [1-E_a^m(\kappa)]dC_a^m(\kappa)dA(a)\right\}}.$$

This is an endogenous variable because young single men decide whether or not to go to school. The strategy here is to pin down $w_{1,1960}$, along with other parameters, such that this statistic is as close as possible to its data counterpart, about 1.55 in 1960.

A similar strategy is followed to determine the gender wage gap parameter ϕ_{1960}. Recall that $\mathbf{M}^f(a,e,a^*,e^*,b,q)$ and $\mathbf{S}^f(a,e)$ are the non-normalized distributions of married and single women, respectively. As in the data, average earnings for

working women are calculated; i.e., all singles and married women who participate in the labor market. This is given by

$$\Big[\phi_{1960} w_{1,1960} \int \cdots \int ae H^f(a,e,a^*,e^*,q) d\mathbf{M}^f(a,e,a^*,e^*,b,q)$$

$$+ \phi_{1960} \int \cdots \int a(1-e) H^f(a,e,a^*,e^*,q) d\mathbf{M}^f(a,e,a^*,e^*,b,q)$$

$$+ \phi_{1960} w_{1,1960} \int\int ae d\mathbf{S}^f(a,e) + \phi_{1960} \int\int a(1-e) d\mathbf{S}^f(a,e)\Big] /$$

$$\Big[\int \cdots \int H^f(a,e,a^*,e^*,q) \mathbf{M}^f(a,e,a^*,e^*,b,q) + \int\int \mathbf{S}^f(a,e)\Big].$$

Again, $w_{0,1960} = 1$. The first and second terms in this equation give the average earnings for married skilled and unskilled women who decide to work. The last two terms calculate the same statistic for single women. Since all men, single or married, work, average earnings for them reads

$$w_{1,1960} \int_0^\infty \int_{-\infty}^\infty a E_a^m(\kappa) dC_a^m(\kappa) dA(a) + \int_0^\infty \int_{-\infty}^\infty a[1 - E_a^m(\kappa)] dC_a^m(\kappa) dA(a).$$

The gender earnings gap in the model is the ratio of these two averages. The parameter ϕ_{1960} is estimated, again along with other parameters, to generate a gender earnings gap in the model that is as close as possible to the observed gender earnings gap in the data, about 0.45 in 1960.

Let DATA represent a vector of 24 moments that are calculated from the US data for 1960. A vector of the analogous 24 moments can be obtained from the steady state of the model for 1960. The moments for the model will be a function of the parameters to be estimated, of course. Therefore, this vector of moments is represented by $\mathcal{M}(\boldsymbol{\omega})$, where $\boldsymbol{\omega}$ denotes the vector of 22 parameters to be estimated. Define the vector of deviations between the data and the model by $\mathbf{G}(\boldsymbol{\omega}) \equiv \text{DATA} - \mathcal{M}(\boldsymbol{\omega})$.

Minimum distance estimation picks the parameter vector, $\boldsymbol{\omega}$, to minimize a weighted sum of the squared deviations between the data and the model. Specifically,

$$\hat{\boldsymbol{\omega}} = \arg\min \mathbf{G}(\boldsymbol{\omega})' \mathbf{W} \mathbf{G}(\boldsymbol{\omega}),$$

where $\mathbf{W}$ is some positive semidefinite matrix. The estimation assumes that the model is a true description of the world, for some value of the parameter vector, $\boldsymbol{\omega}$. The number of targets is larger than the number of parameters. The estimator, $\hat{\boldsymbol{\omega}}$, is consistent for any weighting matrix, $\mathbf{W}$. Let $\text{se}(\hat{\boldsymbol{\omega}})$ represent the vector of standard errors for the estimator, $\hat{\boldsymbol{\omega}}$. It is given by

$$\text{se}(\hat{\boldsymbol{\omega}}) = \text{diag}\left\{\frac{[\mathbf{J}(\hat{\boldsymbol{\omega}})' \mathbf{W} \mathbf{J}(\hat{\boldsymbol{\omega}})]^{-1} \mathbf{J}(\hat{\boldsymbol{\omega}})' \mathbf{W} \boldsymbol{\Sigma} \mathbf{W} \mathbf{J}(\hat{\boldsymbol{\omega}}) [\mathbf{J}(\hat{\boldsymbol{\omega}})' \mathbf{W} \mathbf{J}(\hat{\boldsymbol{\omega}})]^{-1\prime}}{n}\right\},$$

where $\mathbf{J}(\hat{\omega}) \equiv \partial \mathcal{M}(\hat{\omega})/\partial \hat{\omega}$, $\mathbf{\Sigma}$ is the variance-covariance matrix for the data moments, and n is the total number of observations.[22] The data moments are calculated using data from the 1960 US census. Each element in $\mathbf{\Sigma}$ is weighted by the number of observations for a particular moment relative to the total number of observations. Set $\mathbf{W} = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix.

Table 3 reports the parameter estimates and their associated standard errors. The set of moments and the corresponding results for the benchmark model for 1960 are displayed in Table 4. The fitted parameter values look reasonable and are tightly estimated, for the most part.

The estimate of the degree of curvature in the utility function for market goods ($\zeta = 1.78$) is in line with the macroeconomics literature, which typically uses a coefficient of relative aversion of either 1 or 2. Note that nonmarket goods have a weight of $\alpha = 1.20$ in utility. This can be thought of as corresponding to a weight assigned to consumption in a typical macro model of 0.45, with the remaining weight of 0.55 being applied to leisure; i.e., $0.55/0.45 = 1.20$. Nonmarket goods play a role similar to leisure here. Thus, this coefficient does not seem unreasonable. The utility function for nonmarket goods is more concave ($\xi = 3.11$) than the one for market goods. As mentioned in Section III, this implies that a household will tilt its allocation toward market goods as it gets wealthier, and, as a result, this parameter affects the differences in marriage and divorce rates for educated and non-educated individuals.

A household spends about 19 percent of its market consumption on covering the fixed costs of a home (when $\mathfrak{c} = 0.068$). This fixed cost provides an economic motive for marriage since married agents can pool resources to cover $\mathfrak{c}$. It also gives an incentive for married women to participate in the market. If $\mathfrak{c}$ were set to zero, with all other parameters kept at their benchmark values, the fraction of single individuals would be 20 percent (instead of 15 percent). Furthermore, married women are less likely to participate in the labor market. Married female labor-force participation would be only 3.5 percent. The parameters of the marital bliss shocks determine marriage and divorce rates in the model. Note that the distribution for singles has a lower mean (-1.497 versus -0.403) but a higher variance (0.599 versus 0.338) than the one for married couples. This creates an incentive for singles to wait for a match with high b. Once a marriage is formed, marital bliss is quite persistent ($\rho_{b,m} = 0.959$).[23] An educated person realizes 1.308 utils (μ_1) from marrying a similarly educated person. The extra utility for a marriage between two non-educated individuals is lower, 0.4. These are higher than the mean level of bliss in a marriage of -0.403 and influence the level of marital sorting. Setting μ_0 and μ_1 to zero in the 1960 economy would generate a correlation between the education levels for husbands and wives that is close to zero.

The estimation requires that joint work is costly for households with a noncollege-educated husband ($q_l^0 = 0.175$ and $q_h^0 = 0.303$), but there is a benefit

[22] Each diagonal element of $\mathbf{\Sigma}$ corresponds to the variance of a particular moment in the data. Since most moments are calculated with different sample restrictions, off-diagonal terms are set to zero.

[23] In the simulations, the distributions for bliss, as represented by $N(\bar{b}_s, \sigma_{b,s}^2)$ and $b' = (1 - \rho_{b,m})\bar{b}_m + \rho_{b,m} b + \sigma_{b,m}\sqrt{1 - \rho_{b,m}^2}\,\varepsilon$, are approximated on a discrete grid of size 15 using Tauchen's (1986) procedure. Similarly, the distribution for ability, $N(0, \sigma_a^2)$, is approximated on a grid of size 40.

TABLE 3—PARAMETERS: ESTIMATED (*Minimum distance*)

Category	Parameter values	Standard error	95 percent conf. interval
Preferences	$\alpha = 1.198$	0.029	[1.141, 1.255]
	$\xi = 3.114$	0.021	[3.073, 3.155]
	$\zeta = 1.782$	0.010	[1.762, 1.803]
	$\mathfrak{c} = 0.068$	0.0004	[0.067, 0.069]
	$\mu_0 = 0.400$	0.170	[0.067, 0.733]
	$\mu_1 = 1.308$	0.094	[1.124, 1.492]
Ability shocks	$\sigma_a^2 = 0.310$	0.003	[0.304, 0.315]
Marital bliss shocks	$\bar{b}_s = -1.497$	0.111	[−1.715, −1.279]
	$\sigma_{b,s}^2 = 0.599$	0.075	[0.451, 0.746]
	$\bar{b}_m = -0.403$	0.029	[−0.459, −0.347]
	$\sigma_{b,m}^2 = 0.338$	0.028	[0.284, 0.393]
	$\rho_{b,m} = 0.959$	0.004	[0.951, 0.967]
Home shocks	$q_l^0 = 0.175$	0.066	[0.046, 0.305]
	$q_h^0 = 0.303$	0.127	[0.053, 0.552]
	$q_l^1 = -0.226$	0.066	[−0.354, −0.097]
	$q_h^1 = -0.126$	0.123	[−0.367, 0.115]
Price and wages	$p_{1960} = 54.703$	8.219	[38.594, 70.812]
	$w_{0,1960} = 1$ (normalization)	—	—
	$w_{1,1960} = 1.040$	0.015	[1.011, 1.068]
	$\phi_{1960} = 0.400$	0.002	[0.396, 0.404]
Cost of education	$\eta_m = 69.861$	5.525	[59.031, 80.690]
	$\eta_f = 134.970$	8.770	[117.781, 152.159]
	$\sigma_\kappa = 54.134$	4.871	[44.587, 63.681]

of joint work for households in which the husband is educated ($q_l^1 = -0.226$ and $q_h^1 = -0.126$). Given the husband's educational attainment, these parameters determine how the labor-force participation of a married woman changes with her own education. This allows the benchmark economy to produce the observed response of female labor-force participation with respect to female educational attainment. Finally, the variance of the ability distribution, together with the parameters that determine the cost of education, weigh on both the fraction of individuals who choose to get a college education and the overall level of inequality.

As Table 4 illustrates, the model has no problem matching most of the targets. Single women relative to married couples are poorer in the model than they are in the data. The model misses the relative income of households that are composed of a college-educated wife and a noncollege-educated husband. Note, however, that there is a very small number of such households (only 2.8 percent of all marriages). The model yields a slightly higher level of divorce in 1960; 3.3 percent in the data versus 4 percent in the model for college-educated people and 5.3 percent in the data versus 4.4 percent in the model for noncollege-educated ones. As a result, the proportion of singles in the model is also higher than the data in 1960. The model has some difficulty mimicking the very high rate of marriage for noncollege-educated people in 1960.

TABLE 4—DATA AND BENCHMARK MODEL, 1960

	Data		*Model*	
Education	Fem.	Males	Fem.	Males
	0.072	0.125	0.074	0.129
Marriage				
Fraction	Sing.	Marr.	Sing.	Marr.
	0.130	0.870	0.151	0.849
Rates	< Coll.	Coll.	< Coll.	Coll.
Marriage	0.925	0.849	0.888	0.882
Divorce	0.053	0.033	0.044	0.040
Sorting	Wife		Wife	
Husband	< Coll.	Coll.	< Coll.	Coll.
< College	0.855	0.023	0.843	0.028
College	0.082	0.041	0.085	0.045
Corr., educ.	0.414		0.403	
Work, Married Fem.				
Husband	Wife		Wife	
	< Coll.	Coll.	< Coll.	Coll.
< College	0.328	0.528	0.318	0.586
College	0.213	0.347	0.207	0.294
Participation, all	0.324		0.315	
Income, fraction	0.110		0.122	
Inequality				
Gini	0.306		0.307	
Ratio 90/10	4.829		4.536	
Ratio 90/50	1.817		2.043	
Income, Sf/Married	0.473		0.393	
Income, Marriage				
Husband	Wife		Wife	
	< Coll.	Coll.	< Coll.	Coll.
< College	0.932	1.335	0.943	0.700
College	1.369	1.501	1.400	1.501
Skill premium	1.548		1.565	
Gender gap	0.446		0.419	

V. Moving Forward to 2005

The model economy is now ready to be simulated for 2005. This is done using the 2005 prices for durable goods and 2005 wages. As will be seen, in order to match the US data as best as possible, a *very limited* number of parameters need to be retuned for 2005. These parameters involve the utility cost of education and compatibility between individuals of different education levels. There are two key goals of the analysis. The first is to assess the importance of the two driving forces for: the rise in assortative mating; the decline in marriage and the increase in divorce, which has affected noncollege-educated individuals more than college-educated ones; the rise in educational attainments and married female labor-force participation; and the increase in income inequality among households. This assessment is undertaken in

Section VI. Before doing this, it is important for the model to match the US data for 2005. The second goal is to understand the role that the change in family structure plays in generating income inequality. This is done in Section VII. Again, a good fit is desirable before pursuing this goal.

A. *US Stylized Facts and Benchmark Model Results*

To simulate the model economy for 2005, first set $w_{0,2005}$ (the wage rate for unskilled individuals) to 1.17, as the earnings of noncollege-educated men grew by 17 percent between 1960 and 2005. Next, $w_{1,2005}$ (the wage rate for an efficiency unit of skilled labor) and ϕ_{2005} (the gender wage gap) are chosen such that the skill premium and the gender earnings gap in the model economy are as close to their data counterparts as possible. The skill premium increased from 1.55 to 2.02 between 1960 and 2005. At the same time, women's earnings relative to men's increased from 0.45 to 0.64. Matching these two targets in 2005 implies $w_{1,2005} = 1.81$ (versus $w_{1,1960} = 1.04$) and $\phi_{2005} = 0.59$ (versus $\phi_{1960} = 0.40$).

Durable goods were also cheaper in 2005 than they were in 1960. Gordon (1990) reports that the quality-adjusted price of consumer durables declined between 6 percent and 13 percent a year for different durables between 1950 and 1985. A price index for eight durables (refrigerators, air conditioners, washing machines, clothes dryers, TV sets, dishwashers, microwaves, and VCRs) fell at 10 percent a year. In the National Income and Product Accounts (NIPA), the price index for "furnishings and durable household equipment" relative to the price index for "personal consumption expenditures" dropped by about 60 percent between 1960 and 2005 (close to 2 percent a year).[24] In the simulation, it will be assumed that the price of durables falls by 5 percent a year, a value between these two estimates. Consumer durable goods prices in 2005 are then given by $p_{2005} = p_{1960} \times e^{-0.05(2005-1960)}$.[25]

Finally, η_f and η_m are allowed to take different values in 2005. (Recall that given a, an individual of gender g draws κ, the utility cost of an education, from a normal distribution with mean η_g/a and variance σ_κ^2). The 2005 values for these parameters are selected such that the model economy generates exactly the increase in educational attainment that is observed in the data. If these parameters are not allowed to change between 1960 and 2005, the model still generates an increase in the educational attainment, but the increase is smaller, especially so for women.[26] Matching the observed skill premium and the gender earnings gap in the 2005 economy is possible, only if the model also delivers the correct levels of educational attainments for men and women. To match the rise in educational attainment, η_f and η_m had to

[24] The source is NIPA, Table 2.3.4, Price Indexes for Personal Consumption Expenditures by Major Type of Product, version October 30, 2014.

[25] The results for the 2005 model economy (the prelude in Table 5) with lower (2.5 percent) and higher (7.5 percent) price declines are reported in the online Appendix. The decline in marriages and the rise in female labor-force participation are weaker (stronger) with a lower (higher) price decline.

[26] The online Appendix presents the results for the 2005 prelude when η_f and η_m are kept at their 1960 levels. The fraction of men and women who choose a college education would be 20.4 percent and 10.3 percent, respectively. For men, this is about 40 percent of the increase in educational attainment between 1960 and 2005. For women, however, the increase is much smaller. The educational attainment of women would only increase from 7.4 percent to 10.3 percent between 1960 and 2005, which is just 11 percent of observed rise.

be decreased from 134.97 to 66.45 and from 69.86 to 55.75 between the 1960 and 2005 steady states, respectively. The model requires a larger decline in the cost of education for women.[27] All other parameters are kept at their 1960 values.[28] Table 5 shows the results.

Overall, the model does a good job matching the set of stylized facts presented for 2005. First, marriage became less important during this period. Specifically, the fraction of the population that is single more than doubled in the data (from 13.0 to 33.9 percent). The model is able to generate about 40 percent of this increase (15.1 to 23.9 percent). The rise in the number of single people and the fall in the fraction of married individuals is due to both a decline in the rate of marriage and an increase in the rate of divorce. This feature of the data is also matched. The model does deliver a more pronounced decrease in the marriage rates between 1960 and 2005 for noncollege-educated people compared with the college-educated. However, marriage rates for less educated people decline by 6 percentage points in the model (compared with 12 in the data), whereas the decline for college-educated individuals is 5.2 percentage points (and 5.8 in the data). In the data, the increase in the divorce rate is greater for noncollege-educated individuals (5.3 percent to 20.2 percent) vis-à-vis college-educated ones (3.3 percent to 11.9 percent). The model also generates the differential increase in divorce, but the differential increase is less pronounced in the model than it is in the data. The fraction of divorced people increases by 4.9 percentage points for noncollege-educated people (versus 14.9 in the data) and only by 2.0 for college-educated ones (compared with the 8.6 that was observed).

Second, the model does a great job replicating the increase in labor-force participation by married women (from 32.4 to 70.1 percent in the data and 31.5 to 71.6 percent in the model). The model also explains well the upward movement in the share of family income that working wives provide (11.0 to 27.8 percent in the data versus 12.2 to 32.3 percent for the model).

Third, there is more income inequality among households in 2005, both in the data and the model. The Gini coefficient increases from 0.306 to 0.429 between 1960 and 2005 in the data. The model is able to generate about 45 percent of this increase (from 0.307 to 0.362).

[27] Several changes that are not modeled here might be behind these exogenous shifts in education costs. For example: The federal government began guaranteeing student loans in 1965, which increased accessibility to colleges. Moreover, Title IX of the Education Amendments, passed in 1972, banned discrimination against women in education. Another factor might be changes in social norms, which are not explicitly modeled within the current framework.

[28] It is assumed that the survival probability, $1 - \delta$, takes the same value in 1960 and 2005. Life expectancy at birth increased by 7.7 years between 1960 and 2005 (U.S. Census Bureau 2012.) Individuals enter the model economy, however, at age 25 and leave the model at age 55. As a result, the effect of changes in life expectancy for the model economy will be very small. Nevertheless, a counterfactual that adds 7.7 years to life expectancy was conducted, and the results are similar to the benchmark.

TABLE 5—MOVING FORWARD TO 2005 (*A prelude*)

	1960				2005			
	Data		*Model*		*Data*		*Model*	
Education	Fem.	Males	Fem.	Males	Fem.	Males	Fem.	Males
	0.072	0.125	0.074	0.129	0.332	0.318	0.331	0.318
Marriage								
Fraction	Sing.	Marr.	Sing.	Marr.	Sing.	Marr.	Sing.	Marr.
	0.130	0.870	0.151	0.849	0.339	0.661	0.239	0.761
Rates	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
Marriage	0.925	0.849	0.888	0.882	0.806	0.791	0.828	0.830
Divorce	0.053	0.033	0.044	0.040	0.202	0.119	0.093	0.060
Sorting	Wife		Wife		Wife		Wife	
Husband	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.855	0.023	0.843	0.028	0.545	0.108	0.644	0.024
Coll.	0.082	0.041	0.085	0.045	0.109	0.237	0.024	0.308
Corr. educ.; γ	0.414; —		0.403; —		0.519; 0.219		0.892; 0.580	
Work, *Marr. Fem.*								
Husband	Wife		Wife		Wife		Wife	
	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.328	0.528	0.318	0.586	0.685	0.823	0.745	0.440
Coll.	0.213	0.347	0.207	0.294	0.632	0.711	0.793	0.671
Participation, all	0.324		0.315		0.701		0.716	
Income, frac.	0.110		0.122		0.278		0.323	
Inequality								
Gini	0.306		0.307		0.429		0.362	
Ratio 90/10	4.829		4.536		8.219		6.341	
Ratio 90/50	1.817		2.043		2.500		2.359	
Income, Sf/M.	0.473		0.393		0.397		0.391	
Income, Marr.								
Husband	Wife		Wife		Wife		Wife	
	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.932	1.335	0.943	0.700	0.724	1.047	0.737	0.843
Coll.	1.369	1.501	1.400	1.501	1.167	1.534	1.198	1.546
Skill premium	1.548		1.565		2.016		2.014	
Gender gap	0.446		0.419		0.636		0.634	

Finally, the framework has no trouble generating a rise in assortative mating. In fact, the mechanism in the model is too strong. The correlation between a husband's and wife's education increases to 0.892 in the 2005 model economy, whereas it is 0.519 in the 2005 data. As it was highlighted in Section II, the rise in assortative mating can also be captured by the following regression:

$$EDUCATION_t^w = \alpha + \beta \times EDUCATION_t^h + \gamma \times EDUCATION_t^h \times DUMMY_{2005,t} + \theta \times DUMMY_{2005,t}, \tag{16}$$

where $t \in \{1960, 2005\}$. Now, it is shown in Appendix B that it is possible to estimate the parameters of this regression from the information contained in the 2×2 contingency tables for 1960 and 2005. The estimated value of γ, which captures the increase in assortative mating between 1960 and 2005 in the data, is 0.219. In contrast, estimated value for the model economy is almost twice as high at 0.580. Basically, the model has difficulty generating mixed marriages between skilled and unskilled individuals. In particular, there has been a rise in the data for marriages between skilled women and unskilled men, from 2.3 percent of all marriages in 1960 to 10.8 percent of all marriages in 2005.[29] The model economy is not able to generate this increase. The lack of mixed marriages in 2005 also affects how the model economy performs with respect to the labor-force participation of women. In particular, skilled women who are married to skilled men work too little in the model economy compared with what they do in the data, while unskilled women married to unskilled men work too much.

Can the model economy deliver a lower level of sorting in 2005? Consider the following thought experiment: Imagine that the extra utilities of a match between two equally skilled individuals, μ_0 and μ_1, take lower values in 2005. In particular, lower these two parameters such that the contingency table and the coefficient γ in equation (16) are as close as possible to their data counterparts. This requires μ_0 to be reduced from 0.4 to 0.214 and μ_1 from 1.308 to 0.375. One interpretation for this exogenous change is that people are less class conscious today versus yesteryear. The *rise* in positive assortative mating obtained in the model, therefore, comes *solely* from powerful economic forces. The results of this experiment are shown in Table 6.[30] Note that, now, the γ coefficients obtained from running regression (16) both in the data and in the model are much closer (0.219 and 0.216, respectively).[31] Most of the remaining outcomes in Table 6 are very similar to those in Table 5. The 2005 model economy in Table 6 is, however, able to generate a larger decline in the fraction of the married population than the one in Table 5 (from 0.85 to 0.72, instead of from 0.85 to 0.76). The 2005 model economy in Table 6 also does a much better job capturing how married female labor-force participation changes as a function of their own and their husbands' education levels. There is one drawback, however. The model economy in 2005 is not able to generate the differential in divorce rates between skilled and unskilled individuals. Indeed, skilled individuals have a slightly higher divorce rate than unskilled ones in the 2005 model economy (0.123 versus 0.117). All in all, the fit in Table 6 is very good. Take this as the benchmark economy for the subsequent analysis.

[29] Coles and Francesconi (2011) study the emergence of these "toyboy" marriages within a model where individuals value both the wage as well as fitness of their partners.

[30] For the results in Table 6, the calibrated values for skilled wages and the gender gap in 2005 are $w_{1,2005} = 1.875$ and $\phi_{2005} = 0.596$. The cost of education is also slightly altered in this economy to match the observed education rates for men and women. The cost parameters are set to $\eta_{m,2005} = 52.00$ and $\eta_{f,2005} = 55.50$.

[31] The estimated values for the other coefficients in the regression, α, β, and θ, are 0.026, 0.302, and 0.139 in the data, and 0.032, 0.312, and 0.124 in the simulation.

Table 6—Data and Benchmark Model, 1960 and 2005

	1960				2005			
	Data		*Model*		*Data*		*Model*	
Education	Fem.	Males	Fem.	Males	Fem.	Males	Fem.	Males
	0.072	0.125	0.074	0.129	0.332	0.318	0.334	0.317
Marriage								
Fraction	Sing.	Marr.	Sing.	Marr.	Sing.	Marr.	Sing.	Marr.
	0.130	0.870	0.151	0.849	0.339	0.661	0.284	0.716
Rates	<Coll.	Coll.	<Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
Marriage	0.925	0.849	0.888	0.882	0.806	0.791	0.815	0.808
Divorce	0.053	0.033	0.044	0.040	0.202	0.119	0.117	0.123
Sorting	Wife		Wife		Wife		Wife	
Husband	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.855	0.023	0.843	0.028	0.545	0.108	0.574	0.106
Coll.	0.082	0.041	0.085	0.045	0.109	0.237	0.101	0.219
Corr. educ.; γ	0.414; —		0.403; —		0.519; 0.219		0.526; 0.216	
Work, Marr. Fem.								
Husband	Wife		Wife		Wife		Wife	
	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.328	0.528	0.318	0.586	0.685	0.823	0.742	0.823
Coll.	0.213	0.347	0.207	0.294	0.632	0.711	0.577	0.795
Participation, all	0.324		0.315		0.701		0.745	
Income, frac.	0.110		0.122		0.278		0.335	
Inequality								
Gini	0.306		0.307		0.429		0.366	
Ratio 90/10	4.829		4.536		8.219		6.214	
Ratio 90/50	1.817		2.043		2.500		2.348	
Income, Sf/M.	0.473		0.393		0.397		0.418	
Income, Marr.								
Husband	Wife		Wife		Wife		Wife	
	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.932	1.335	0.943	0.700	0.724	1.047	0.727	0.884
Coll.	1.369	1.501	1.400	1.501	1.167	1.534	1.313	1.629
Skill premium	1.548		1.565		2.016		2.019	
Gender gap	0.446		0.419		0.636		0.635	

VI. Under the Hood

The forces underlying the decline in marriage, the increase in assortative mating, the upswing in married female labor-force participation, the rise in educational attainment, and higher income inequality will now be inspected. These forces are labor-saving technological progress in the home, a rise in the general level of wages, a widening in the college premium, and a narrowing of the gender wage gap. Two experiments are considered here. First, technological advance in the household sector will be shut down. Hence, only the structure of wages changes in this experiment. Second, shifts in the wage structure are turned off. Now, there is only technological

progress in the home. The analysis takes the results in Table 6 as the benchmark for the two experiments.

A. *No Technological Progress in the Home* (*Change in the Wage Structure Only*)

To begin with, consider shutting down technological progress in the home. Thus, only changes in the wage structure (the general level of wages, the skill premium and the gender wage gap) are operational. Specifically, fix the 2005 price of household inputs, p, at the 1960 level. All other parameters are set at the values used to produce the 2005 benchmark model economy presented in Table 6. Think about this experiment as representing a comparative statics exercise, one done numerically as opposed to the more traditional qualitative analysis that uses pencil and paper techniques. The results of this experiment are shown in Table 7. As can be seen from the table, technological progress in the household sector is vital for promoting married female labor-force participation. Without it, very few married women, about 26 percent, would work in 2005. In fact, a slightly lower fraction of educated women would work in 2005 than in 1960. This is because households are richer in 2005 than in 1960, because of a rise in wages.[32]

Producing home goods is labor intensive. Married households are better disposed to undertake household production relative to single ones because they have a larger endowment of time. Hence, the lack of technological progress in the home makes marriage more attractive. In the benchmark economy, the number of married individuals declines from 85 percent to 72 percent between 1960 and 2005, a drop of 13 percentage points. The decline is smaller without technological progress in the home. The number of married individuals is now about 78 percent in 2005, a decline of about 7 percentage points. This decline is due to higher wages and a lower gender wage gap, which make singlehood more affordable. A higher skill premium makes skilled individuals choosier in the marriage market, and consequently, it boosts the degree of assortative mating. The rise in assortative mating is, however, smaller than the benchmark economy: The correlation between a husband's and wife's education increases from 0.40 to 0.44. A similar conclusion can be drawn by comparing the corresponding γ coefficients from the regression in (16): γ decreases from 0.216 in the 2005 benchmark to 0.088 in this counterfactual. When women do not work, the upward movement in the skill premium has a smaller effect on marital sorting since, in this case, their wage is not important. Finally, income inequality in 2005 remains roughly constant when technological progress in the home is eliminated.

B. *No Change in Wage Structure* (*Technological Progress in the Home Only*)

Now consider the situation in which there is only technological progress in the home; i.e., shut down changes in wages. In particular, set wages for both men and

[32] The absence of technological progress in the home leads to a large drop in married female labor supply. One might think that the equilibrium level of wages will rise in response. This could operate to dampen the withdrawal of labor effort by women. The structure employed here assumes that production is linear in male and female work effort, so such an effect is precluded. This assumption is relaxed in the online Appendix; the results are similar.

TABLE 7—NO TECHNOLOGICAL PROGRESS IN THE HOME (*Change in wage structure only*)

	1960		2005			
	Benchmark		Experiment		Benchmark	
Education	Fem.	Males	Fem.	Males	Fem.	Males
	0.074	0.129	0.275	0.348	0.334	0.317
Marriage						
Fraction	Sing.	Marr.	Sing.	Marr.	Sing.	Marr.
	0.151	0.849	0.221	0.779	0.284	0.716
Rates	<Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
Marriage	0.888	0.882	0.826	0.854	0.815	0.808
Divorce	0.044	0.040	0.067	0.067	0.117	0.123
Sorting	Wife		Wife		Wife	
Husband	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.843	0.028	0.555	0.070	0.574	0.106
Coll.	0.085	0.045	0.183	0.192	0.101	0.219
Corr. educ.; γ	0.403; —		0.441; 0.088		0.526; 0.216	
Work, *Marr. Fem.*						
Husband	Wife		Wife		Wife	
	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.318	0.586	0.207	0.237	0.742	0.823
Coll.	0.207	0.294	0.244	0.449	0.577	0.795
Participation, all	0.315		0.262		0.745	
Income, frac.	0.122		0.129		0.335	
Inequality						
Gini	0.307		0.363		0.366	
Ratio 90/10	4.536		6.002		6.214	
Ratio 90/50	2.043		2.494		2.348	
Income, Sf/M.	0.393		0.507		0.418	
Income, Marr.						
Husband	Wife		Wife		Wife	
	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.943	0.700	0.712	0.969	0.727	0.884
Coll.	1.400	1.501	1.294	1.563	1.313	1.629
Skill premium	1.565		1.869		2.019	
Gender gap	0.419		0.721		0.635	

women at the levels they had in 1960; i.e., $w_{0,2005} = w_{0,1960}$, $w_{1,2005} = w_{1,1960}$, and $\phi_{2005} = \phi_{1960}$.[33] The results of this comparative statics experiment are shown in Table 8. Observe first that the fraction of married women who work in 2005 is now 64 percent. This is only 10.5 percentage points less than the number of married women who work in the 2005 benchmark economy (74.5 percent). Therefore,

[33] The results when only the gender wage gap is kept at its 1960 value are shown in the online Appendix. The importance of the narrowing gender gap for changes in married female labor-force participation is stressed by Jones, Manuelli, and McGrattan (2015).

TABLE 8—NO CHANGE IN WAGE STRUCTURE (*Technological progress in the home only*)

	1960		2005			
	Benchmark		Experiment		Benchmark	
Education	Fem.	Males	Fem.	Males	Fem.	Males
	0.074	0.129	0.220	0.194	0.334	0.317
Marriage						
Fraction	Sing.	Marr.	Sing.	Marr.	Sing.	Marr.
	0.151	0.849	0.255	0.745	0.284	0.716
Rates	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
Marriage	0.888	0.882	0.835	0.820	0.815	0.808
Divorce	0.044	0.040	0.103	0.108	0.117	0.123
Sorting	Wife		Wife		Wife	
Husband	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.843	0.028	0.693	0.114	0.574	0.106
Coll.	0.085	0.045	0.094	0.099	0.101	0.219
Corr. educ.; γ	0.403; —		0.356; 0.057		0.526; 0.216	
Work, Marr. Fem.						
Husband	Wife		Wife		Wife	
	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.318	0.586	0.634	0.761	0.742	0.823
Coll.	0.207	0.294	0.613	0.565	0.577	0.795
Participation, all	0.315		0.640		0.745	
Income, frac.	0.122		0.263		0.335	
Inequality						
Gini	0.307		0.333		0.366	
Ratio 90/10	4.536		6.529		6.214	
Ratio 90/50	2.043		1.919		2.348	
Income, Sf/M.	0.393		0.281		0.418	
Income, Marr.						
Husband	Wife		Wife		Wife	
	< Coll.	Coll.	< Coll.	Coll.	< Coll.	Coll.
< Coll.	0.943	0.700	0.916	0.954	0.727	0.884
Coll.	1.400	1.501	1.267	1.391	1.313	1.629
Skill premium	1.565		1.439		2.019	
Gender gap	0.419		0.438		0.635	

growth in wages is not the key driver of the rise in married female labor-force participation. Technological progress in the household sector is.

Marriage still declines significantly, from 85 percent to 75 percent. This decline of 10 percentage points is about three-quarters of the total 13 percentage point decline between 1960 and 2005. Hence, while both advancement in wages and home technologies affect marriage and divorce decisions, the effect of home technologies is relatively more important. In contrast, without changes in wages, the degree of assortative mating remains more or less constant, as can be seen from the changes in γ (it decreases substantially from 0.216 in the benchmark to 0.057 in this

experiment). Hence, wages are key for shifts in assortative mating. Furthermore, without the growth in wages, the hike in inequality is smaller.

VII. Households and Inequality

Income inequality among households increased between 1960 to 2005. How much of this upturn is driven by changes in wages? And how much of it is because of the propagation mechanisms stressed here: the decisions of households regarding education, marriage, and married female labor supply? To address this question, take the 1960 economy and change wages (the skill premium and the gender wage gap) and durable goods prices to their 2005 values.[34] Though prices are changing, consider keeping the decisions regarding education, marriage, and married female labor-force participation constant at their 1960 values. With these modified prices and artificially fixed decisions, a new counterfactual steady state can be computed. Calculate the Gini coefficient for this hypothetical scenario. In this experiment, if an individual was not going to college in the 1960 economy, he/she still chooses not to go to college, despite a higher skill premium. If a woman decided to get married, she still does so, even though household technology and the gender wage gap have improved. Note that since all decisions are fixed in their 1960 levels, the lower price of durables has no effect other than allowing individuals to enjoy a higher utility, due to the positive income effect. As a result, the outcome of this experiment shows how much shifts in wages, per se, contribute to the hike in inequality.

The results are shown in column 2 in Table 9. Column 1 reports the Gini coefficient for the 1960 benchmark model economy. The Gini coefficient increases from 0.307 to 0.330. This constitutes 39 percent of the total increase in the Gini, from 0.307 to 0.366. So, shifts in wages are clearly an important driver of the hike in income inequality. Still, the model's propagation mechanism is very important, accounting for the remaining 61 percent. This propagation mechanism will be examined now. To do this, redo the previous experiment but now allow households to adjust the labor-force participation decisions for married women. Education and marriage decisions are still kept at their 1960 values. Married female labor-force participation rises from 31.5 percent in the 1960 benchmark to 61.6 percent in this counterfactual economy because of cheaper consumer durables. The Gini coefficient, however, does not change, as seen in column 3. Changes in female labor-force participation alone do not affect inequality. Next, keep married female labor-force participation decisions at their 1960 values and let marriage decisions change. The results are shown in column 4 of Table 9. When marriage decisions are allowed to react, the number of married individuals declines from 0.85 percent to 0.68 percent. Though the degree of positive assortative mating slightly decreases in this counterfactual experiment, this still results in higher inequality.[35] This is due to the fact that

[34] The wage structure, education costs and compatibility parameters, μ_0 and μ_1, imposed here are all taken from the 2005 economy described in Table 6.

[35] The spousal education correlation decreases from 0.40 in the 1960 benchmark to 0.34 in this counterfactual. Remember that here the extra utilities derived when two equally skilled individuals marry, μ_0 and μ_1, are set to

Table 9—Deconstructing the Increase in Household Income Inequality

	1960	Experiments				2005
	(1)	(2)	(3)	(4)	(5)	(6)
Decisions held fixed		Edu., Marr. LFP	Edu., Marr.	Edu., LFP	Edu.	
Gini	**0.307**	**0.330**	**0.330**	**0.341**	**0.351**	**0.366**
Change in Gini	0.000	0.023	0.023	0.034	0.044	0.059
Cumulative change	**0.0%**	**39.0%**	**39.0%**	**57.6%**	**74.6%**	**100.0%**

Notes: The Gini in the benchmark model increases between the 1960 and 2005 steady states. The table breaks this shift down into a series of steps. The table starts with the 1960 benchmark economy in column 1. Column 2 shows what would happen if only wages were allowed to change between 1960 and 2005; i.e., the education, marriage and divorce, and married female labor-force participation decisions are all held fixed at the 1960 levels. Column 3 allows only married female labor-force participation decisions to adjust to 2005 wages and prices. A similar experiment for the marriage and divorce decisions is undertaken in column 4, where all other decisions are held fixed at 1960 levels. Column 5 allows both participation and marriage decisions to change; i.e., only education decisions are held fixed. Column 6 shows the move to the 2005 benchmark economy; hence, now education is free to move in addition to the other decisions.

single households are much poorer than married ones, especially for women—recall the facts presented in Table 6. Marriage decisions account for about 18.6 percent (57.6 percent minus 39 percent) of the rise in income inequality.

In column 5 of Table 9, both marriage and labor-force participation decisions are allowed to adjust. Education decisions are still kept in their 1960 values. The level of inequality moves up even further. Observe the nonlinear interaction effect. Allowing only female labor-force participation to adjust had no effect on inequality. Likewise, permitting just the marriage decisions to respond accounted for 18.6 percent of the changes in inequality. But allowing female labor-force participation and marriage decisions to react *together* accounts for 35.6 percent (74.6 minus 39 percent) of the total climb. The effect of changes in marriage patterns (who is married, who is single, and who marries with whom) is magnified when married women are allowed to adjust their labor-force participation. A rise in the skill premium and a reduction in the gender wage gap boost the tendency toward positive assortative mating. For this effect to be fully operational, married women must work. A skilled man is indifferent on economic grounds between a skilled and unskilled woman if neither of them works, assuming that skill doesn't effect a woman's production value at home. When both work, however, the skilled woman becomes the more attractive partner, at least from an economic point of view.

Finally, the gap between columns 5 and 6 shows the contribution of endogenous education, and the subsequent induced changes in marriage and married female labor-force participation decisions, on income inequality. Not surprisingly, allowing education decisions to respond hikes income inequality. When the skill premium rises, more high-ability people will go to school. This amplifies the spread between what high-ability and low-ability people earn.

lower values relative to the 1960 benchmark. Thus, the modest decrease in this correlation somewhat conceals the powerful forces behind the rise in positive assortative mating.

VIII. Conclusions

People today are more likely to marry someone of the same socioeconomic class than they were in the past. At the same time, the prevalence of marriage has fallen and the occurrence of divorce has risen, especially for people without a college education. Women are much more likely to go to college now. Married ones work more. Household income inequality has intensified. This has led to a dramatic transformation of the American household.

To address these facts, a model of marriage and divorce is developed. In the constructed framework, individuals marry for both economic and noneconomic reasons. The noneconomic reasons are companionship and love. The economic ones are the values of a spouse's labor at home and in the market. Technological progress in the household sector erodes the value of labor in home production, thus reducing the importance of such labor in a marriage. As a result, married women enter the labor market. Love becomes a more important determinant in marriage. An individual can now afford to delay marriage and wait to find a mate that makes him or her happy. This leads to a decline in marriage and a rise in divorce, which contribute to the growth in income inequality. Increases in the college premium provide an incentive for both young men and women to go to college. A college-educated person earns more in both married and single life. The fact that men now desire women who make a good income provides an extra incentive for a young woman to go to college, or vice versa. An additional motivation may be that people like to marry others with the same educational background. In equilibrium, this leads to a rise in assortative mating.

The structural model developed is fit to US data using a minimum distance estimation procedure. A collection of data moments summarizing educational attainment, the patterns of marriage and divorce, married female labor-force participation, and income inequality in 1960 is targeted. The estimated structural model matches the stylized facts well, yielding parameter values that are both reasonable and tightly estimated. The model's predictions for 2005 are also broadly in line with the data. Like almost everything in life there is still room for improvement. In particular, the model generates too steep an increase in assortative mating. Before deconstructing the effects of technological progress in the home and changes in the wage structure on the variables of interest, a small set of parameters are retuned to generate a reasonable 2005 benchmark.

The decomposition exercises show that technological progress in the home is an important factor for explaining the rise in married female labor-force participation. The narrowing of the gender wage gap plays an ancillary role here. Technological progress in the home is also a significant driver of the decline in marriage and rise in divorce. The structure of wages in the US has a powerful influence on assortative mating and educational attainment. As the skill premium climbs, income inequality widens. This increase is magnified by the endogenous forces at work: higher levels of educational attainment, stronger positive assortative mating, and the hike in married female labor-force participation magnify the rise in household income inequality.

Appendix

A. *Data*

Unless stated otherwise, all data is obtained from the Integrated Public Use Microdata Series-USA (IPUMS-USA). For the years 1960, 1970, 1980, 1990, and 2000, the data derives from federal censuses, while for 2005 it comes from the American Community Survey (ACS). The ACS has a sample size comparable to the one percent census samples that the IPUMS-USA provides for the other years. The age group for which the analysis is done is 25–54. Only singles and married couples are considered. Widows, widowers and married individuals whose spouses are absent are excluded from the analysis. The wage variable is restricted to be nonnegative. Furthermore, all single female and male households in which the household head does not work or has zero wages are excluded. All married households in which the male earner does not work or have zero wages are excluded too. These restrictions are motivated by the economic environment used in the paper. By following this procedure, the moments for data are the exact counterparts to those arising from the model. These moments are used in the estimation of the model. A college-educated individual refers to someone with four years of college or more, otherwise the person is labeled as being noncollege-educated. This applies to both men and women.

Figure 1.—The fraction of the population that is ever married is one minus the fraction of the population that is never married. The fraction of the population that is currently divorced is calculated by taking the stock of currently divorced and then dividing it by the stock of ever-married people.

Figure 2.—The value of γ_t is plotted from the regression equation (1). This equation is estimated for married couples using the data mentioned above. The regression coefficient measures the incremental likelihood (relative to 1960) that an educated man is married to an educated woman in the year t, for $t = 1970, 1980, 1990, 2000,$ and 2005.

Figure 3.—Married female labor-force participation is calculated from the variable employee status (EMPSTAT) in IPUMS. This variable takes one of three values: working, not working, and not in the labor force. A woman is assumed to be in the labor force if EMPSTAT $= 1$; i.e., if she is working. This calculation is done for both college and noncollege-educated women. A wife's contribution to family income is calculated by computing the ratio of her labor income to total family labor income. This ratio is averaged across all married women.

Figure 4.—A woman is labeled as having a college degree if she has four years of college or more. The college premium is calculated by dividing average labor income for college-educated men by average labor income for noncollege-educated ones. The gender wage gap is calculated as the ratio of the average wage for working women to the average wage for working men.

Figure 5.—Single and married households are sorted in an increasing order by their total household labor income. The Lorenz curves and the Gini coefficients are computed based on this ordering.

B. *Fitting a Linear Regression Model to the Contingency Tables*

Consider running the following regression for the years 1960 and 2005:

$$\begin{aligned} \textit{EDUCATION}_t^w = \alpha + \beta \times \textit{EDUCATION}_t^h + \gamma \times \textit{EDUCATION}_t^h \times \textit{DUMMY}_t^{05} \\ + \theta \times \textit{DUMMY}_t^{05}, \end{aligned}$$

where $\textit{EDUCATION}_t^w \in \{0, 1\}$ is the observed level of the wife's education in period $t = 1960, 2005$ and takes a value of one, if the woman completed college, and a value of zero, otherwise; $\textit{EDUCATION}_t^h \in \{0, 1\}$ is the husband's education; $\textit{DUMMY}_t^{05}$ is a dummy variable for time such that $\textit{DUMMY}_t^{05} = 1$, if $t = 2005$, and $\textit{DUMMY}_t^{05} = 0$, if $t = 1960$. The coefficient γ measures the additional impact relative to 1960 that a husband's education will have on his wife's in 2005. On the other hand, denote the contingency tables for 1960 and 2005 by

$$\begin{bmatrix} p_{<c,<c}^{60} & p_{<c,c}^{60} \\ p_{c,<c}^{60} & p_{c,c}^{60} \end{bmatrix} \text{ and } \begin{bmatrix} p_{<c,<c}^{05} & p_{<c,c}^{05} \\ p_{c,<c}^{05} & p_{c,c}^{05} \end{bmatrix}.$$

The rows give the husband's education levels, the columns give the wife's education levels. The elements in the contingency table give the population moments for each of the four types of marriages for the two years in question.

To map the contingency tables into the regression, pick the four parameters α, β, γ, and θ to solve the following least squares minimization problem, which minimizes the prediction error for the regression across the four types of marriage in each of the two years:

$$\begin{aligned} \min_{\alpha,\beta,\theta,\gamma} \big\{ & p_{<c,<c}^{60}(-\alpha)^2 + p_{<c,c}^{60}(1-\alpha)^2 + p_{c,<c}^{60}(-\alpha-\beta)^2 + p_{c,c}^{60}(1-\alpha-\beta)^2 \\ & + p_{<c,<c}^{05}(-\alpha-\theta)^2 + p_{<c,c}^{05}(1-\alpha-\theta)^2 \\ & + p_{c,<c}^{05}(-\alpha-\beta-\gamma-\theta)^2 + p_{c,c}^{05}(1-\alpha-\beta-\gamma-\theta)^2 \big\}. \end{aligned}$$

To understand why, focus on the first term, which represents a type-$(< c, < c)$ marriage in 1960. This occurs with odds $p_{<c,<c}^{60}$. Plug the education level for the husband, or $\textit{EDUCATION}_{1960}^h = 0$, into the regression equation. The regression equation predicts an answer of α. But, $\textit{EDUCATION}_t^w = 0$ when the wife has a less than college education. So, the term $(0 - \alpha)^2 = (-\alpha)^2$ is the square of the prediction error for a

type- $(< c, \ < c)$ marriage in 1960. The first-order conditions associated with this problem are represented by a system of four linear equations:

$$p^{60}_{<c,<c}\alpha - p^{60}_{<c,c}(1-\alpha) - p^{60}_{c,<c}(-\alpha-\beta) - p^{60}_{c,c}(1-\alpha-\beta)$$

$$- p^{05}_{<c,<c}(-\alpha-\theta) - p^{05}_{<c,c}(1-\alpha-\theta) - p^{05}_{c,<c}(-\alpha-\beta-\gamma-\theta) - p^{05}_{c,c}(1-\alpha-\beta-\gamma-\theta) = 0,$$

$$- p^{60}_{c,<c}(-\alpha-\beta) - p^{60}_{c,c}(1-\alpha-\beta) - p^{05}_{c,<c}(-\alpha-\beta-\gamma-\theta) - p^{05}_{c,c}(1-\alpha-\beta-\gamma-\theta) = 0,$$

$$- p^{05}_{<c,<c}(-\alpha-\theta) - p^{05}_{<c,c}(1-\alpha-\theta) - p^{05}_{c,<c}(-\alpha-\beta-\gamma-\theta) - p^{05}_{c,c}(1-\alpha-\beta-\gamma-\theta) = 0,$$

and

$$-p^{05}_{c,<c}(-\alpha-\beta-\gamma-\theta) - p^{05}_{c,c}(1-\alpha-\beta-\gamma-\theta) = 0.$$

The solution to this system of linear equations is

$$\alpha = \frac{p^{60}_{<c,c}}{p^{60}_{<c,<c} + p^{60}_{<c,c}}, \qquad \beta = \frac{p^{60}_{c,c}}{p^{60}_{c,<c} + p^{60}_{c,c}} - \alpha,$$

$$\theta = \frac{p^{05}_{<c,c}}{p^{05}_{<c,<c} + p^{05}_{<c,c}} - \alpha, \qquad \gamma = \frac{p^{05}_{c,c}}{p^{05}_{c,<c} + p^{05}_{c,c}} - \alpha - \beta - \theta.$$

The results are:

	Regression coefficients			
	α	β	γ	θ
Data	0.026	0.302	0.219	0.139
Benchmark model	0.032	0.312	0.216	0.124

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